

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-14

Contents

Action Center	2
Full Sample PC1	3
Position Sheet	4
Cluster Overview	6
Cluster Details	7
Long ETF Positions (full)	59
ETF Net Signal Strength	70
Diversifier Candidates	72
Run Parameters	73

Action Center**Run date:** 2026-05-14 **Universe:** 661 ETFs / 26 clusters**Full PC1:** LONG since 2026-04-13 (D+22, 3.7%) **Clusters long:** 17/26 **ETFs long:** 464/661 (70%)**Today's events (cluster level)**

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

SELL (2)

cluster	theme	spread_sell	p_h5	nss
cluster_13	Fixed Income	-0.209	0.96	-0.791
cluster_7	Aggregate Fixed Income	-0.107	0.97	-0.392

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 4 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_22	Cash / Ultra-Short Fixed Income	0.393	1	1	1.000	1.00
cluster_19	Japan Equity	9.476	1	1	1.000	0.99
cluster_1	Developed ex-US Equity	4.609	1	1	0.962	0.53
cluster_16	Municipal Bond	0.346	1	1	0.132	0.04

Full Sample PC1

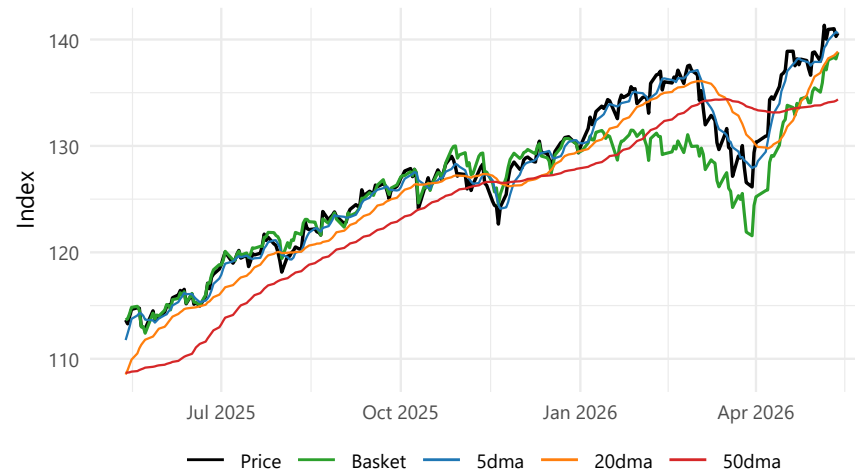


Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-04-13 (22 days, 3.7%)

5dma: 140.576 20dma: 138.867 50dma: 134.368

Spread vs 50dma: +6.208 Spread vs 20dma: +1.708

P(buy crossover within 5 days): 100.0% P(sell): 11.9%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
SRLN	cluster_21	Bank Loan / Floating Rate	1.000	26	1.2%	6.22%	0.89
FLOT	cluster_21	Bank Loan / Floating Rate	1.000	26	1.2%	5.84%	0.89
BKLN	cluster_21	Bank Loan / Floating Rate	1.000	26	1.2%	4.18%	0.89
QQQ	cluster_4	U.S. Large Cap Growth	1.000	22	14.7%	3.37%	0.97
FLRN	cluster_21	Bank Loan / Floating Rate	1.000	26	1.2%	2.55%	0.89
FLTR	cluster_21	Bank Loan / Floating Rate	1.000	26	1.2%	0.68%	0.89
IEMG	cluster_8	Emerging Markets Equity	0.999	23	8.7%	2.68%	0.99
VOO	cluster_5	U.S. Large Cap Equity	0.983	22	6.2%	4.82%	0.99
PDBC	cluster_20	Broad Commodities	0.946	86	49.5%	2.45%	0.98
KWEB	cluster_12	China Equity	0.939	20	4.5%	2.03%	0.92
GUNR	cluster_9	Materials	0.874	26	4.8%	3.37%	0.96
AMLP	cluster_9	Materials	0.874	26	4.8%	0.55%	0.96
URA	cluster_9	Materials	0.874	26	4.8%	0.34%	0.96
IAU	cluster_15	Gold	0.787	1	-0.0%	1.08%	0.95
SLV	cluster_15	Gold	0.787	1	-0.0%	0.72%	0.95
GLD	cluster_15	Gold	0.787	1	-0.0%	0.20%	0.95
BWX	cluster_24	International Fixed Income	0.714	9	-0.4%	7.46%	0.93
VNQ	cluster_14	Real Estate	0.637	23	3.5%	4.95%	0.99
VTIP	cluster_18	Inflation-Protected Fixed I...	0.627	24	0.7%	34.47%	0.90
VTV	cluster_2	U.S. Large Cap Value	0.495	22	2.1%	6.99%	0.98
XLP	cluster_23	Consumer Staples	0.488	8	-0.8%	5.12%	0.97
IJH	cluster_3	U.S. Mid Cap Equity	0.462	22	2.6%	3.62%	0.99
HYG	cluster_10	High Yield Bond	0.225	22	0.4%	11.62%	0.95
VHT	cluster_11	Healthcare	0.006	5	-0.4%	4.58%	0.93
IJR	cluster_6	U.S. Small Cap Equity	0.005	23	1.8%	4.28%	0.98

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
VEA	cluster_1	Developed ex-US Equity	1	0.526	0.922	0.99

EWJ	cluster_19	Japan Equity	1	0.991	0.908	0.96
MUB	cluster_16	Municipal Bond	1	0.044	0.905	0.95
ICSH	cluster_22	Cash / Ultra-Short Fixed In...	1	1.000	0.419	0.84
SHV	cluster_22	Cash / Ultra-Short Fixed In...	1	1.000	0.349	0.84
MINT	cluster_22	Cash / Ultra-Short Fixed In...	1	1.000	0.175	0.84
GBIL	cluster_22	Cash / Ultra-Short Fixed In...	1	1.000	0.117	0.84

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
4	cluster_4	U.S. Large Cap Growth	U.S. Large Cap Gr...	62	LONG	shifted	22	14.7%	1.000	+0.009
21	cluster_21	Bank Loan / Floating Rate	Bank Loan / Float...	9	LONG	shifted	26	1.2%	1.000	+0.092
22	cluster_22	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT	shifted			1.000	+0.000
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	LONG		23	8.7%	0.999	-0.001
19	cluster_19	Japan Equity	Japan Equity	9	FLAT	shifted			0.991	+0.247
5	cluster_5	U.S. Large Cap Equity	U.S. Large Cap Eq...	57	LONG	shifted	22	6.2%	0.983	+1.214
20	cluster_20	Broad Commodities	Broad Commodities	9	LONG	shifted	86	49.5%	0.946	-0.034
12	cluster_12	China Equity	China Equity	15	LONG	shifted	20	4.5%	0.939	+0.203
9	cluster_9	Materials	Materials	22	LONG	shifted	26	4.8%	0.874	+0.442
15	cluster_15	Gold	Gold	13	LONG		1	-0.0%	0.787	+0.214
24	cluster_24	International Fixed Income	International Fix...	6	LONG		9	-0.4%	0.714	-0.171
14	cluster_14	Real Estate	Real Estate	14	LONG		23	3.5%	0.637	-0.193
18	cluster_18	Inflation-Protected Fixed I...	Inflation-Protect...	10	LONG	shifted	24	0.7%	0.627	+0.594
1	cluster_1	Developed ex-US Equity	Developed ex-US E...	82	FLAT				0.526	-0.124
2	cluster_2	U.S. Large Cap Value	U.S. Large Cap Va...	71	LONG	shifted	22	2.1%	0.495	-0.280
23	cluster_23	Consumer Staples	Consumer Staples	7	LONG	shifted	8	-0.8%	0.488	-0.512
3	cluster_3	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	66	LONG	shifted	22	2.6%	0.462	-0.538
10	cluster_10	High Yield Bond	High Yield Bond	21	LONG	shifted	22	0.4%	0.225	-0.318
16	cluster_16	Municipal Bond	Municipal Bond	13	FLAT				0.044	-0.130
11	cluster_11	Healthcare	Healthcare	17	LONG		5	-0.4%	0.006	-0.043
6	cluster_6	U.S. Small Cap Equity	U.S. Small Cap Eq...	48	LONG	shifted	23	1.8%	0.005	-0.871
17	cluster_17	India Equity	India Equity	12	FLAT	shifted			-0.275	+0.275
7	cluster_7	Aggregate Fixed Income	Aggregate Fixed I...	43	FLAT	shifted			-0.392	-0.488
13	cluster_13	Fixed Income	Fixed Income	15	FLAT	shifted			-0.791	-1.703
25	cluster_25	Utilities	Utilities	4	FLAT	shifted			-1.000	-0.952
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				-1.000	-0.000

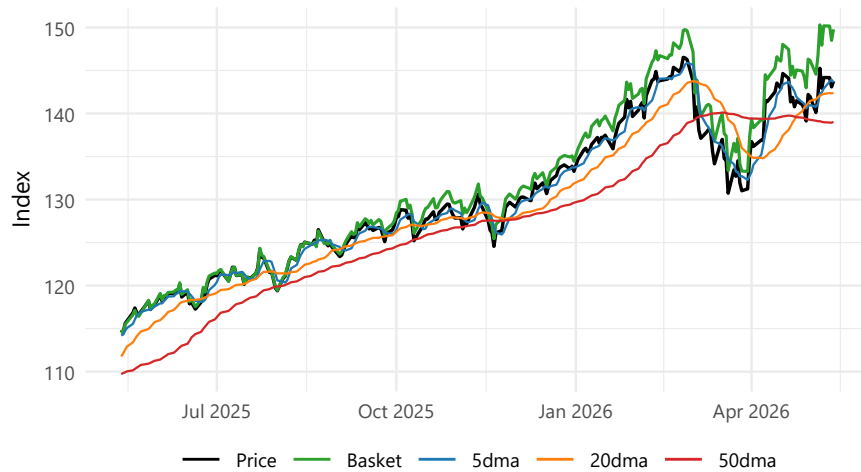
Cluster Details

Cluster 1 – Developed ex-US Equity

Members: 82 **Var explained:** 0.839 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +44.8%)
- Latest sell_cross: 2026-05-04
- Current 252d return: +25.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
International Equity	21	0.2686
Developed ex-US Equity	9	0.1179
International Dividend Equity	6	0.0748
Global ex-US Equity	5	0.0656
Europe Equity	4	0.0518
Alternatives	4	0.0464
Eurozone Equity	3	0.0372
Asia-Pacific Equity	2	0.0242

Real Estate	2	0.0225
Global Equity	2	0.0222
Emerging Markets Fixed Income	2	0.0193
Currency	2	0.0181
Commodity	1	0.0132
Unknown	1	0.0130
Global ex-US Small Cap Equity	1	0.0129
Germany Equity	1	0.0124
France Equity	1	0.0124
United Kingdom Equity	1	0.0123
Italy Equity	1	0.0122
Europe Financials	1	0.0120
Sweden Equity	1	0.0120
Netherlands Equity	1	0.0120
Australia Equity	1	0.0119
Spain Equity	1	0.0116
Canada Equity	1	0.0114
Switzerland Equity	1	0.0114
Belgium Equity	1	0.0113
Austria Equity	1	0.0112
Poland Equity	1	0.0104
South Africa Equity	1	0.0099
New Zealand Equity	1	0.0091
Mexico Equity	1	0.0089

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
VEA	0.922	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0133	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0133	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
63	IEFA	0.0133	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
64	SCHF	0.0133	Schwab International Equity ETF	Developed ex-US Equity	LONG
65	SPDW	0.0133	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
21	DWM	0.0132	WisdomTree International Equity Fund	International Dividend E...	FLAT
34	TLTD	0.0132	FlexShares Morningstar Developed Market...	International Equity	FLAT
49	GSIE	0.0132	Goldman Sachs ActiveBeta International ...	Commodity	FLAT

52	ESGD	0.0132	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
28	PXF	0.0131	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
38	ACWX	0.0131	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
39	CWI	0.0131	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
41	INTF	0.0131	iShares International Equity Factor ETF	International Equity	FLAT
42	IQLT	0.0131	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
44	QEFA	0.0131	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
58	IEUR	0.0131	iShares Core MSCI Europe ETF	Europe Equity	FLAT
59	IXUS	0.0131	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
60	VEU	0.0131	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
61	VGK	0.0131	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
62	VXUS	0.0131	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
6	IEV	0.0130	iShares Europe ETF	Europe Equity	FLAT
32	EFV	0.0130	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
35	JPIN	0.0130	JPMorgan Diversified Return Internation...	International Equity	FLAT
46	RODM	0.0130	Hartford Multifactor Developed Markets ...	International Equity	FLAT
47	VYMI	0.0130	Vanguard International High Dividend Yi...	International Dividend E...	LONG
48	FNDF	0.0130	Schwab Fundamental International Equity...	International Equity	LONG
56	SCHC	0.0130	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
75	FEU	0.0130			FLAT
24	IQDF	0.0129	FlexShares International Quality Divide...	International Equity	LONG
31	SCZ	0.0129	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
33	FNDC	0.0129	Schwab Fundamental International Small ...	International Equity	LONG
57	VSS	0.0129	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
8	DLS	0.0128	WisdomTree International SmallCap Fund	International Equity	LONG
19	EZU	0.0128	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
43	IVLU	0.0128	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
72	HFXI	0.0128	NYLI FTSE International Equity Currency...	International Equity	LONG
29	EFG	0.0127	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
45	FEZ	0.0127	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
54	VIGI	0.0127	Vanguard International Dividend Appreci...	International Dividend E...	LONG
4	FEP	0.0126	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
30	GWX	0.0126	State Street SPDR S&P International Sma...	International Equity	LONG
40	IMTM	0.0126	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
1	EWG	0.0124	iShares MSCI Germany Index Fund	Germany Equity	LONG
5	VIDI	0.0124	Vident International Equity Strategy ETF	International Equity	LONG
16	EWQ	0.0124	iShares MSCI France Index Fund	France Equity	FLAT
20	IDV	0.0124	iShares International Select Dividend E...	International Dividend E...	LONG
53	HDEF	0.0124	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
17	EWU	0.0123	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
50	IDLV	0.0123	Invesco S&P International Developed Low...	International Equity	LONG
79	IPKW	0.0123	Invesco International BuyBack Achievers...	International Equity	LONG
82	EWI	0.0122	iShares MSCI Italy ETF	Italy Equity	LONG
22	EPP	0.0121	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
51	EFAV	0.0121	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT

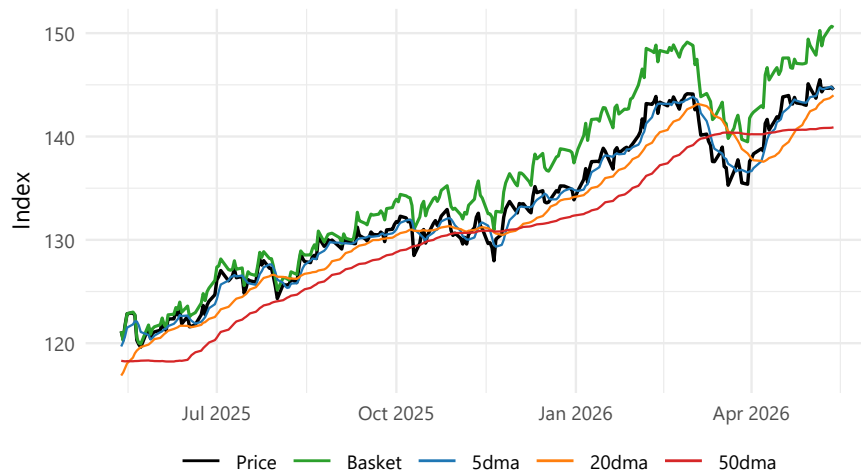
23	EUFN	0.0120	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
71	EWD	0.0120	iShares MSCI Sweden ETF	Sweden Equity	LONG
77	EWN	0.0120	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
81	BBAX	0.0120	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
13	EWA	0.0119	iShares MSCI Australia Index Fund	Australia Equity	FLAT
27	DWX	0.0119	State Street SPDR S&P International Div...	International Dividend E...	LONG
12	HEZU	0.0118	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
36	DBEF	0.0118	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
37	HEFA	0.0118	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
11	PID	0.0117	Invesco International Dividend Achiever...	International Dividend E...	LONG
15	EWP	0.0116	iShares MSCI Spain ETF	Spain Equity	FLAT
9	HEDJ	0.0114	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
10	IHDG	0.0114	WisdomTree International Hedged Quality...	Alternatives	LONG
14	EWL	0.0114	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
18	EWK	0.0114	iShares MSCI Canada Index Fund	Canada Equity	LONG
25	NFRA	0.0114	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
55	VNQI	0.0114	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
80	EWK	0.0113	iShares MSCI Belgium ETF	Belgium Equity	FLAT
76	EWO	0.0112	iShares MSCI Austria ETF	Austria Equity	LONG
7	RWX	0.0111	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
26	IGF	0.0108	iShares Global Infrastructure ETF	Global Equity	LONG
70	EPOL	0.0104	iShares MSCI Poland ETF	Poland Equity	FLAT
74	EZA	0.0099	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
69	FXA	0.0097	Invesco CurrencyShares Australian Dolla...	Currency	LONG
66	EBND	0.0096	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
67	EMLC	0.0096	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
73	ENZL	0.0091	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
68	EWX	0.0089	iShares MSCI Mexico ETF	Mexico Equity	LONG
78	FXC	0.0084	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT

Cluster 2 – U.S. Large Cap Value

Members: 71 **Var explained:** 0.893 **Signal:** LONG since 2026-04-13 (22 days, 2.1%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +23.1%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +19.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	22	0.3075
U.S. Large Cap Value	10	0.1470
Value Equity	4	0.0568
U.S. Low Volatility Equity	4	0.0527
U.S. Multifactor Equity	3	0.0430
U.S. Value Factor Equity	3	0.0426
U.S. Mid Cap Equity	3	0.0423
U.S. Fundamental Equity	2	0.0293
U.S. Mid Cap Value	2	0.0291
U.S. Equal Weight Equity	2	0.0290
U.S. Equity	2	0.0290
Dividend Equity	2	0.0271
U.S. Mega Cap Value	1	0.0145
U.S. Revenue Weighted Equity	1	0.0145
Large Cap Blend Equities	1	0.0144
U.S. Size Factor Equity	1	0.0142
U.S. Buyback Equity	1	0.0141
U.S. Large Cap Equity	1	0.0140
U.S. Quality / Factor Equity	1	0.0138
U.S. Moat Equity	1	0.0135
U.S. Defensive Equity	1	0.0135
U.S. ESG Equity	1	0.0134

Global Equity	1	0.0129
Communication Services	1	0.0120

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.008

ticker	rep_wt	name	category	etf_signal
VTV	1.01	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
26	IWD	0.0148	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
34	VONV	0.0148	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
41	SCHV	0.0148	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
2	RSP	0.0147	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
10	PRF	0.0147	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
29	JPUS	0.0147	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	LONG
38	ILCV	0.0147	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
40	IUSV	0.0147	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
44	VTV	0.0147	Vanguard Value ETF	U.S. Large Cap Value	LONG
61	DTD	0.0147	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
15	CFA	0.0146	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
18	NULV	0.0146	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
22	FNDX	0.0146	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
24	IWX	0.0146	iShares Russell Top 200 Value ETF	Value Equity	LONG
28	IVE	0.0146	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
32	VOOV	0.0146	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
37	VYM	0.0146	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG
42	SPYV	0.0146	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
56	VOE	0.0146	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
60	DLN	0.0146	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
66	DGRO	0.0146	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
11	RWL	0.0145	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
13	AIVL	0.0145	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
25	ONEV	0.0145	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	LONG
35	MGV	0.0145	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
53	IWS	0.0145	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
70	JPME	0.0145	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	LONG
5	FEX	0.0144	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
45	FTA	0.0144	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	FLAT
54	ONEY	0.0144	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG

58	PFM	0.0144	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
23	EQAL	0.0143	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
67	VIG	0.0143	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
17	NOBL	0.0142	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
19	SDY	0.0142	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
20	FDVV	0.0142	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
30	SIZE	0.0142	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
36	SCHD	0.0142	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
39	IMCB	0.0142	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
71	JVAL	0.0142	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
3	FVD	0.0141	First Trust VL Dividend	U.S. Dividend Equity	FLAT
4	PKW	0.0141	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	FLAT
6	PWV	0.0141	Invesco Large Cap Value ETF	Value Equity	LONG
27	IWR	0.0141	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
43	VO	0.0141	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
48	SDOG	0.0141	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
49	DVY	0.0141	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
63	DIA	0.0140	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
7	RDVY	0.0139	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	LONG
31	VLUE	0.0139	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
65	LGLV	0.0139	State Street SPDR US Large Cap Low Volat...	U.S. Low Volatility Equi...	FLAT
55	SPYD	0.0138	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
57	FTCS	0.0138	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	FLAT
59	OUSA	0.0138	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
16	CFO	0.0137	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
51	RPV	0.0137	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
12	DHS	0.0136	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
8	MOAT	0.0135	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	LONG
14	CDC	0.0135	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
64	USMV	0.0135	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
50	RDIV	0.0134	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
68	ETHO	0.0134	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	LONG
52	SPHD	0.0133	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
9	FDL	0.0132	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
46	PEY	0.0131	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	LONG
47	DIV	0.0130	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
1	ACWV	0.0129	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
21	LVHD	0.0129	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
33	HDV	0.0129	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
62	SPLV	0.0124	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT
69	IYZ	0.0120	iShares U.S. Telecommunications ETF	Communication Services	FLAT

Cluster 3 – U.S. Mid Cap Equity

Members: 66 **Var explained:** 0.848 **Signal:** LONG since 2026-04-13 (22 days, 2.6%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +31.7%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +23.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)

**Category breakdown**

category	n_members	abs_weight
Industrials	15	0.2179
U.S. Small Cap Growth	6	0.0954
U.S. Mid Cap Growth	6	0.0950
Consumer Discretionary	6	0.0845
U.S. Mid Cap Equity	5	0.0813
U.S. Small Cap Equity	5	0.0803
U.S. Momentum Equity	3	0.0427
Utilities	3	0.0427
Growth Equity	2	0.0307
U.S. Small Cap Momentum Equity	2	0.0304
Momentum Equity	2	0.0296

Unknown	1	0.0163
U.S. Small Cap ESG Equity	1	0.0162
U.S. Small Cap Multifactor Equity	1	0.0161
U.S. Extended Market Equity	1	0.0161
U.S. Mid Cap Quality Equity	1	0.0158
U.S. Mid Cap Momentum Equity	1	0.0154
U.S. Small Cap Industrials	1	0.0151
Technology	1	0.0150
U.S. Micro Cap Equity	1	0.0150
IPO Equity	1	0.0143
U.S. Small Cap Consumer Discretionary	1	0.0142

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
IJH	1.009	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
23	SCHM	0.0163	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
29	MDY	0.0163	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	LONG
37	IVOO	0.0163	Vanguard S&P Mid-Cap 400 ETF		LONG
42	IJH	0.0163	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG
43	SPMD	0.0163	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	LONG
44	VB	0.0163	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG
20	JHMM	0.0162	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	LONG
26	NUSC	0.0162	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	LONG
31	IJK	0.0162	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
33	IVOG	0.0162	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
34	MDYG	0.0162	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
45	SCHA	0.0162	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
27	SMLF	0.0161	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
41	VXF	0.0161	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
4	IWM	0.0160	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
30	IJT	0.0160	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	LONG
32	VIOG	0.0160	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	LONG
35	SLYG	0.0160	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	LONG
38	VTWO	0.0160	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
40	ISCG	0.0159	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
15	FXR	0.0158	First Trust Industrials AlphaDEX	Industrials	FLAT

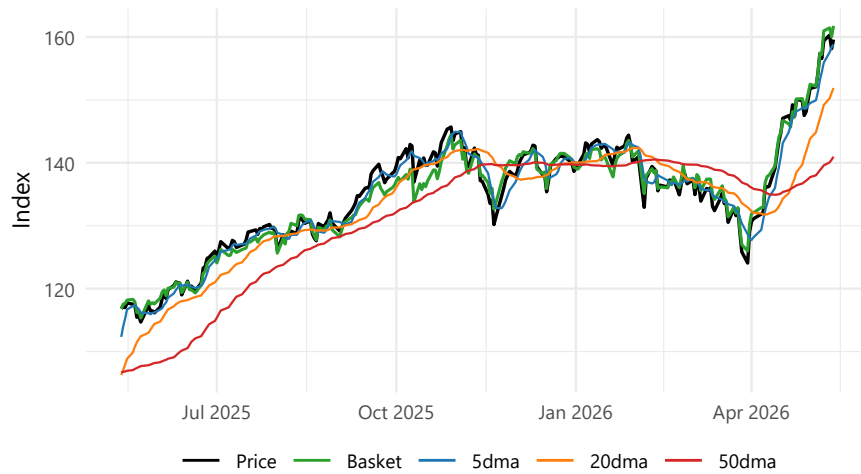
28	IWO	0.0158	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
36	VTWG	0.0158	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
39	VBK	0.0158	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
54	PSC	0.0158	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
61	XMHQ	0.0158	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	LONG
14	IMCG	0.0157	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
21	VIS	0.0157	Vanguard Industrials ETF	Industrials	LONG
19	IYJ	0.0156	iShares U.S. Industrials ETF	Industrials	LONG
22	FIDU	0.0156	Fidelity MSCI Industrials Index ETF	Industrials	LONG
60	PAVE	0.0155	Global X U.S. Infrastructure Developmen...	Industrials	LONG
10	XMMO	0.0154	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
11	IWP	0.0153	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
13	VOT	0.0153	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
46	FXD	0.0153	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
62	VFMO	0.0153	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
66	XSMO	0.0153	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG
1	XLI	0.0152	State Street Industrial Select Sector S...	Industrials	LONG
59	PSCI	0.0151	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	LONG
6	FTC	0.0150	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
24	DWAS	0.0150	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
25	IWC	0.0150	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
53	RSPD	0.0150	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
7	PDP	0.0149	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
9	RPG	0.0148	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
18	FIW	0.0148	First Trust Water ETF	Utilities	LONG
48	FTXR	0.0148	First Trust Nasdaq Transportation ETF	Industrials	LONG
5	FV	0.0147	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
63	PKB	0.0147	Invesco Building & Construction ETF	Industrials	FLAT
64	PRN	0.0146	Invesco Dorsey Wright Industrials Momen...	Industrials	LONG
16	PHO	0.0145	Invesco Water Resources ETF	Utilities	FLAT
8	FPX	0.0143	First Trust US Equity Opportunities ETF	IPO Equity	LONG
57	XTN	0.0143	State Street SPDR S&P Transportation ETF	Industrials	FLAT
47	PEJ	0.0142	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
65	PSCD	0.0142	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
52	IYT	0.0141	iShares U.S. Transportation ETF	Industrials	LONG
2	XRT	0.0139	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
58	QMOM	0.0138	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
12	MTUM	0.0137	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
56	XHB	0.0137	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
17	CGW	0.0135	Invesco S&P Global Water Index ETF	Utilities	FLAT
55	XAR	0.0135	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
49	PPA	0.0133	Invesco Aerospace & Defense ETF	Industrials	FLAT
3	JETS	0.0127	U.S. Global Jets ETF	Industrials	LONG
50	ITA	0.0125	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT
51	ITB	0.0124	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 4 – U.S. Large Cap Growth

Members: 62 **Var explained:** 0.835 **Signal:** LONG since 2026-04-13 (22 days, 14.7%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +34.5%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +39.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
Technology	34	0.5450
U.S. Large Cap Growth	11	0.1891
U.S. Equity	2	0.0339
Growth Equity	2	0.0338
Convertible Bond	2	0.0311
Financials	2	0.0303
U.S. Mega Cap Growth	1	0.0171
U.S. Equal Weight Equity	1	0.0171
U.S. Small Cap Technology	1	0.0156
IPO Equity	1	0.0155
Options Income	1	0.0155

Trend-Following Equity	1	0.0150
Consumer Discretionary	1	0.0150
U.S. Trend-Following Equity	1	0.0131
Israel Equity	1	0.0129

Replication basket – QQQ (cor=0.97) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
QQQ	1.023	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
35	IGM	0.0174	iShares Expanded Tech Sector ETF	Technology	LONG
40	ONEQ	0.0174	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
14	QQQ	0.0173	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
50	ILCG	0.0173	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
29	XNTK	0.0172	State Street SPDR NYSE Technology ETF	Technology	LONG
37	IYW	0.0172	iShares U.S. Technology ETF	Technology	LONG
39	NULG	0.0172	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
43	IWF	0.0172	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
45	VGT	0.0172	Vanguard Information Tech ETF	Technology	LONG
47	FTEC	0.0172	Fidelity MSCI Information Technology In...	Technology	LONG
48	VONG	0.0172	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
52	SCHG	0.0172	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
54	VUG	0.0172	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
8	AIQ	0.0171	Global X Artificial Intelligence & Tech...	Technology	LONG
17	QQEW	0.0171	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
18	QTEC	0.0171	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
24	FXL	0.0171	First Trust Technology AlphaDEX	Technology	LONG
44	IVW	0.0171	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
46	VOOG	0.0171	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
49	MGK	0.0171	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
51	IUSG	0.0171	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
53	SPYG	0.0171	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
1	XLK	0.0170	State Street Technology Select Sector S...	Technology	LONG
36	IXN	0.0170	iShares Global Tech ETF	Technology	LONG
42	IWY	0.0170	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
21	XT	0.0169	iShares Future Exponential Technologies...	Technology	LONG
34	PWB	0.0168	Invesco Large Cap Growth ETF	Growth Equity	LONG
55	SPHB	0.0165	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG

20	TDIV	0.0163	First Trust NASDAQ Technology Dividend ...	Technology	LONG
6	BOTZ	0.0162	Global X Robotics & Artificial Intellig...	Technology	LONG
7	ROBO	0.0162	ROBO Global Robotics and Automation Ind...	Technology	LONG
28	FDN	0.0162	First Trust DJ Internet Index Fund	Technology	LONG
62	QTUM	0.0162	Defiance Quantum ETF	Technology	LONG
27	PNQI	0.0161	Invesco Nasdaq Internet ETF	Technology	FLAT
5	SKYY	0.0160	First Trust Cloud Computing ETF	Technology	LONG
2	SOXX	0.0159	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
3	SMH	0.0159	VanEck Semiconductor ETF	Technology	LONG
4	IGV	0.0158	iShares Expanded Tech-Software Sector E...	Technology	LONG
22	XSD	0.0158	State Street SPDR S&P Semiconductor ETF	Technology	LONG
38	CWB	0.0158	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
13	ARKQ	0.0157	ARK Autonomous Technology & Robotics ETF	Technology	LONG
15	NXTG	0.0157	First Trust Indxx NextG ETF	Technology	LONG
61	PTF	0.0157	Invesco Dorsey Wright Technology Moment...	Technology	LONG
19	PSI	0.0156	Invesco Semiconductors ETF	Technology	LONG
58	PSCT	0.0156	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
26	IPO	0.0155	Renaissance IPO ETF	IPO Equity	LONG
31	FINX	0.0155	Global X FinTech ETF	Financials	LONG
33	QYLD	0.0155	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
60	FTXL	0.0155	First Trust Nasdaq Semiconductor ETF	Technology	LONG
12	ARKW	0.0154	ARK Next Generation Internet ETF	Technology	LONG
41	ICVT	0.0154	iShares Convertible Bond ETF	Convertible Bond	LONG
11	ARKK	0.0151	ARK Innovation ETF	Technology	LONG
25	CIBR	0.0151	First Trust NASDAQ Cybersecurity ETF	Technology	LONG
23	IBUY	0.0150	Amplify Online Retail ETF	Consumer Discretionary	FLAT
32	PTNQ	0.0150	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
10	HACK	0.0149	Amplify Cybersecurity ETF	Technology	LONG
30	IPAY	0.0148	Amplify Digital Payments ETF	Financials	FLAT
16	PRNT	0.0147	3D Printing (The) ETF	Technology	LONG
9	BLOK	0.0140	Amplify Blockchain Technology ETF	Technology	LONG
56	PTLC	0.0131	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
57	EIS	0.0129	iShares MSCI Israel ETF	Israel Equity	LONG
59	BLCN	0.0125	Siren NexGen Economy ETF	Technology	LONG

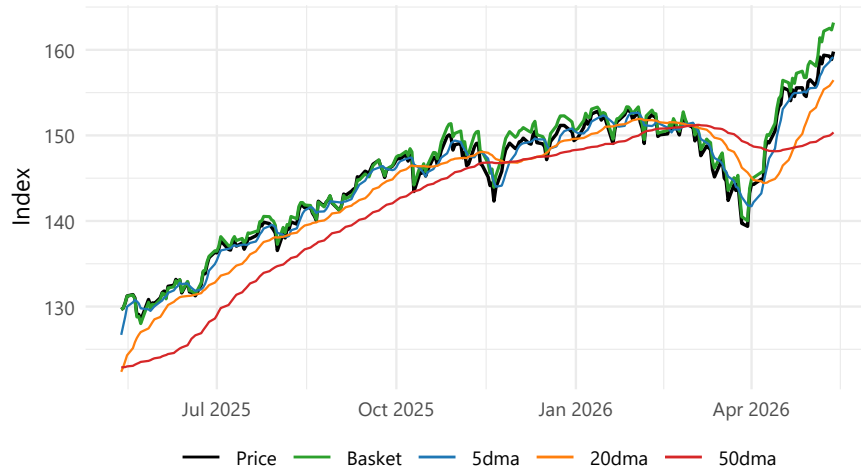
Cluster 5 – U.S. Large Cap Equity

Members: 57 **Var explained:** 0.906 **Signal:** LONG since 2026-04-13 (22 days, 6.2%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.2%)

- Latest sell_cross: 2026-03-06
- Current 252d return: +24.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1653
U.S. Total Market Equity	7	0.1288
Global Equity	7	0.1264
Consumer Discretionary	6	0.0970
Asset Allocation	4	0.0670
U.S. Mega Cap Equity	3	0.0542
U.S. Quality / Factor Equity	3	0.0535
U.S. Multifactor Equity	3	0.0534
Communication Services	3	0.0483
U.S. ESG Equity	2	0.0366
U.S. Equity	2	0.0366
Commodity	1	0.0183
Large Cap Value Equities	1	0.0182
Large Cap Blend Equities	1	0.0182
Dividend Equity	1	0.0178
U.S. Dividend Equity	1	0.0175
Alternatives	1	0.0158
International IPO Equity	1	0.0141

Singapore Equity 1 0.0129

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
VOO	0.958	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	SPY	0.0184	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
30	IWV	0.0184	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
31	IYY	0.0184	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
36	ESGU	0.0184	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
37	IWB	0.0184	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
42	VTHR	0.0184	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
45	VONE	0.0184	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
47	VV	0.0184	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
48	ILCB	0.0184	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
49	ITOT	0.0184	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
50	IVV	0.0184	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
51	SCHB	0.0184	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
52	SCHX	0.0184	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
53	SPTM	0.0184	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
54	VOO	0.0184	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
55	VTI	0.0184	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
12	URTH	0.0183	iShares MSCI World ETF	Global Equity	LONG
29	SUSA	0.0183	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
34	SPYX	0.0183	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
38	IWL	0.0183	iShares Russell Top 200 ETF	U.S. Equity	LONG
43	GSLC	0.0183	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
46	MGC	0.0183	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
7	ACWI	0.0182	iShares MSCI ACWI ETF	Global Equity	LONG
15	VT	0.0182	Vanguard Total World Stock Index ETF	Global Equity	LONG
22	TILT	0.0182	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
26	JHML	0.0182	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
28	DSI	0.0182	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
32	LRGF	0.0182	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
44	EPS	0.0182	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
13	CRBN	0.0181	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
14	SPGM	0.0181	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG

33	OEF	0.0181	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
39	QUAL	0.0181	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
8	AOA	0.0180	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
40	QUS	0.0179	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
23	IOO	0.0178	iShares Global 100 ETF	Global Equity	LONG
24	QDF	0.0178	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
25	CATH	0.0178	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	XLG	0.0178	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
41	SPHQ	0.0176	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
11	AOR	0.0175	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
27	DGRW	0.0175	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
17	IYC	0.0172	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
57	OMFL	0.0171	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
18	VCR	0.0168	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
21	FDIS	0.0168	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
1	XLY	0.0166	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	AOM	0.0164	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
19	VOX	0.0162	Vanguard Communication Services ETF	Communication Services	LONG
20	FCOM	0.0162	Fidelity MSCI Communication Services In...	Communication Services	LONG
2	XLC	0.0158	State Street Communication Services Sel...	Communication Services	LONG
5	QAI	0.0158	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
56	RTH	0.0156	VanEck Retail ETF	Consumer Discretionary	FLAT
9	AOK	0.0150	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
16	FPXI	0.0141	First Trust International Equity Opport...	International IPO Equity	LONG
3	ESPO	0.0140	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
6	EWS	0.0129	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 6 – U.S. Small Cap Equity

Members: 48 **Var explained:** 0.880 **Signal:** LONG since 2026-04-10 (23 days, 1.8%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +35.3%)
- Latest sell_cross: 2026-02-27
- Current 252d return: +17.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-02-27)



Category breakdown

category	n_members	abs_weight
Financials	13	0.2576
U.S. Small Cap Value	6	0.1304
U.S. Small Cap Equity	5	0.1082
U.S. Mid Cap Value	3	0.0657
Dividend Equity	2	0.0424
Leveraged / Inverse	2	0.0422
U.S. Small Cap Low Volatility	2	0.0407
U.S. Small Cap Fundamental Equity	1	0.0218
Mid Cap Value Equities	1	0.0218
U.S. Mid Cap Dividend Equity	1	0.0217
Small Cap Value Equities	1	0.0216
U.S. Small Cap Dividend Equity	1	0.0216
U.S. Mid/Small Cap Equity	1	0.0216
U.S. Small/Mid Cap Fundamental Equity	1	0.0215
Value Equity	1	0.0215
U.S. Mid Cap Equity	1	0.0214
U.S. Small/Mid Cap Equity	1	0.0213
U.S. Equity	1	0.0210
U.S. Dividend Equity	1	0.0206
U.S. Mid Cap Low Volatility	1	0.0198
Global Equity	1	0.0181

Real Estate 1 0.0173

Replication basket – IJR (cor=0.98) basket lead at lag 1: -0.001

ticker	rep_wt	name	category	etf_signal
IJR	0.904	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
37	VBR	0.0220	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	LONG
28	IJJ	0.0219	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
30	IVOV	0.0219	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
31	MDYV	0.0219	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
21	EZM	0.0218	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	LONG
23	FNDA	0.0218	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	LONG
8	FYX	0.0217	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	LONG
10	VIOO	0.0217	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	LONG
11	IJR	0.0217	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG
12	SPSM	0.0217	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	LONG
19	DON	0.0217	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	LONG
26	IWN	0.0217	iShares Russell 2000 Value ETF	U.S. Small Cap Value	LONG
29	IJS	0.0217	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
32	SLYV	0.0217	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	LONG
34	VTWV	0.0217	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	LONG
47	SDVY	0.0217	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	FLAT
7	FNX	0.0216	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	LONG
18	DES	0.0216	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	LONG
20	EES	0.0216	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	LONG
33	VIOV	0.0216	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	LONG
9	PRFZ	0.0215	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	LONG
46	FYT	0.0215	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
41	RWK	0.0214	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	LONG
38	RWJ	0.0213	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	LONG
48	ISMD	0.0213	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
39	RFV	0.0212	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	LONG
42	OMFS	0.0210	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	LONG
13	FXO	0.0209	First Trust Financials AlphaDEX	Financials	FLAT
43	RZV	0.0209	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
6	REGL	0.0207	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	FLAT
17	SMDV	0.0206	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT

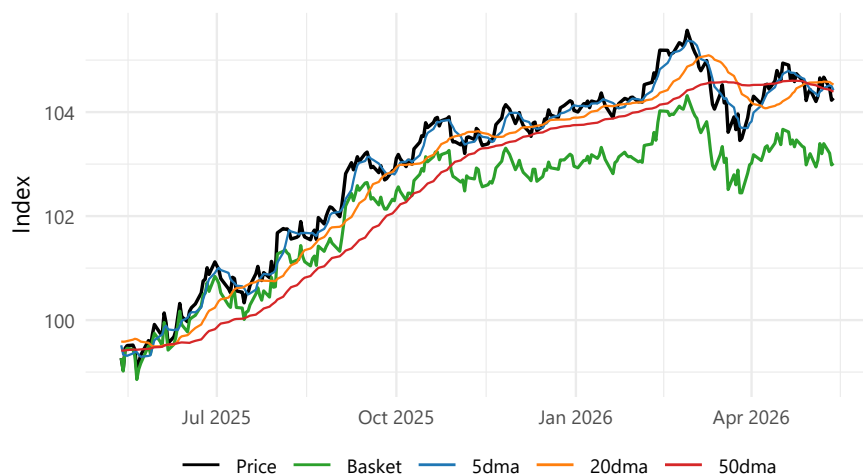
25	XSLV	0.0204	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
35	VFH	0.0204	Vanguard Financials ETF	Financials	FLAT
36	FNCL	0.0204	Fidelity MSCI Financials Index ETF	Financials	LONG
3	KBE	0.0203	State Street SPDR S&P Bank ETF	Financials	FLAT
27	SMMV	0.0203	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
15	IYF	0.0202	iShares U.S. Financial ETF	Financials	FLAT
45	KCE	0.0202	State Street SPDR S&P Capital Markets E...	Financials	LONG
16	IYG	0.0201	iShares U.S. Financial Services ETF	Financials	FLAT
24	XMLV	0.0198	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
1	XLF	0.0197	State Street Financial Select Sector SP...	Financials	LONG
22	KBWB	0.0197	Invesco KBW Bank ETF	Financials	FLAT
2	KRE	0.0196	State Street SPDR S&P Regional Banking ...	Financials	FLAT
14	IAT	0.0194	iShares U.S. Regional Banks ETF	Financials	FLAT
44	IAI	0.0189	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
40	PSP	0.0181	Invesco Global Listed Private Equity ETF	Global Equity	LONG
4	KIE	0.0177	State Street SPDR S&P Insurance ETF	Financials	FLAT
5	REM	0.0173	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 7 – Aggregate Fixed Income

Members: 43 **Var explained:** 0.842 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-08 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +5.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-08)



Category breakdown

category	n_members	abs_weight
Fixed Income	13	0.3080
Aggregate Fixed Income	6	0.1478
Investment Grade Corporate Bond	6	0.1440
Mortgage-Backed Fixed Income	6	0.1326
Active Fixed Income	4	0.0943
Cash / Ultra-Short Fixed Income	3	0.0667
Target Maturity Bond	2	0.0423
International Fixed Income	2	0.0417
Agency Fixed Income	1	0.0226

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
BND	0.728	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

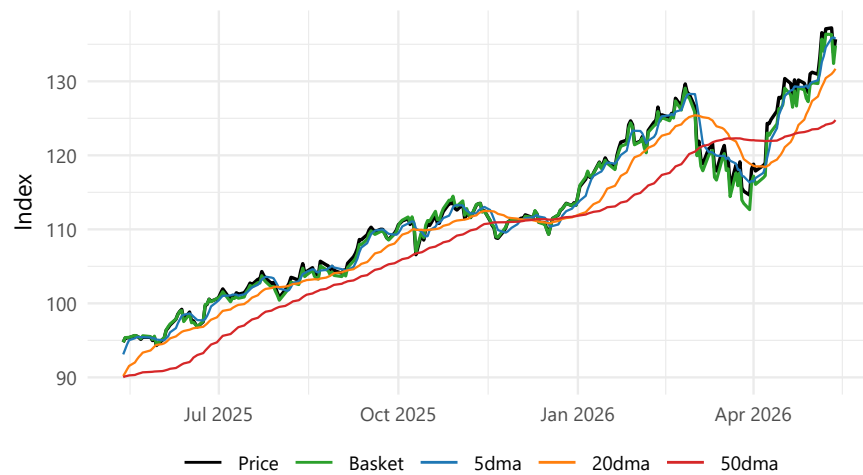
	ticker	PC_wt	name	category	etf_signal
13	BIV	0.0250	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
16	AGG	0.0249	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
33	VGIT	0.0249	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
17	BND	0.0248	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
19	SCHZ	0.0248	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
29	IEI	0.0248	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
30	SPTI	0.0248	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
32	SCHR	0.0248	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
12	IUSB	0.0247	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
20	SPAB	0.0247	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
1	IEF	0.0246	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
6	GVI	0.0246	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
11	ISTB	0.0245	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
9	SPIB	0.0243	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	LONG
10	IGIB	0.0243	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	LONG
14	BSV	0.0242	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
15	VCIT	0.0242	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	LONG
35	MBB	0.0242	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
36	VMBS	0.0242	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
5	FBND	0.0241	Fidelity Total Bond ETF	Active Fixed Income	FLAT
37	SPMB	0.0241	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
24	IGSB	0.0240	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
4	GTO	0.0239	Invesco Total Return Bond ETF	Active Fixed Income	LONG
7	AGGY	0.0239	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
26	VCSH	0.0239	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
3	BOND	0.0237	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
25	SLQD	0.0234	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
31	GOVT	0.0233	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
34	GNMA	0.0233	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
38	SHY	0.0228	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
23	SPSB	0.0226	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
28	AGZ	0.0226	iShares Agency Bond ETF	Agency Fixed Income	LONG
18	TOTL	0.0225	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
41	VGSH	0.0225	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
40	SCHO	0.0221	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
39	SPTS	0.0220	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
22	BSCQ	0.0215	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
42	BNDX	0.0210	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
8	IBDR	0.0208	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
43	IAGG	0.0207	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
27	CMBS	0.0186	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	LONG
2	LMBS	0.0181	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
21	LDUR	0.0171	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.874 **Signal:** LONG since 2026-04-10 (23 days, 8.7%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +43.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7724
Asia Equity	2	0.0812
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0357
Global Dividend Equity	1	0.0347
South Korea Equity	1	0.0341

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.025

ticker	rep_wt	name	category	etf_signal
IEMG	0.93	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0419	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
10	AIA	0.0401	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0390	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0386	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0357	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0347	Global X SuperDividend ETF	Global Dividend Equity	LONG
25	FEMS	0.0346	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	LONG

Cluster 9 – Materials

Members: 22 **Var explained:** 0.677 **Signal:** LONG since 2026-04-07 (26 days, 4.8%)

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +43.0%)

- Latest sell_cross: 2026-03-09
- Current 252d return: +45.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-03-09)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6459
MLP / Energy Infrastructure	4	0.1700
Energy	1	0.0497
Consumer Staples	1	0.0488
Norway Equity	1	0.0436
Industrial Metals	1	0.0328
Natural Gas	1	0.0092

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.044

ticker	rep_wt	name	category	etf_signal
AMLP	0.124	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.078	Global X Uranium ETF	Materials	LONG
GUNR	0.764	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

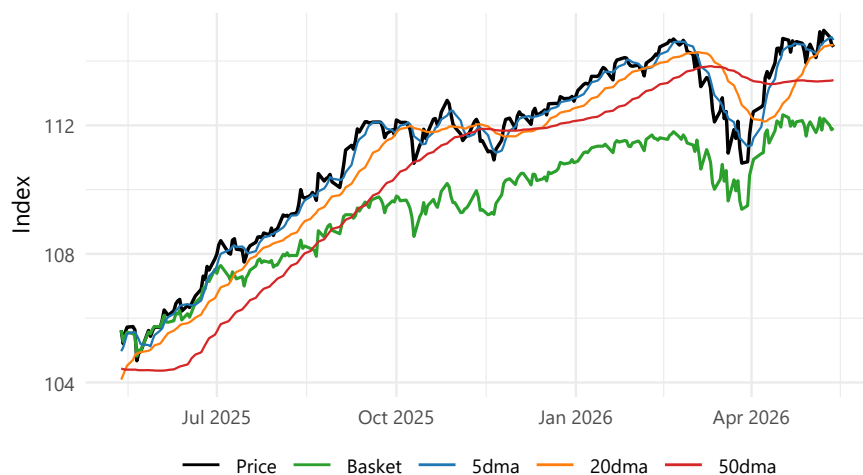
	ticker	PC_wt	name	category	etf_signal
15	GNR	0.0543	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0536	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0522	iShares Global Materials ETF	Materials	FLAT
11	IYM	0.0522	iShares U.S. Basic Materials ETF	Materials	FLAT
22	NANR	0.0518	State Street SPDR S&P North American Na...	Materials	LONG
13	VAW	0.0507	Vanguard Materials ETF	Materials	FLAT
16	PICK	0.0502	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0500	Invesco S&P 500 Equal Weight Materials ...	Materials	FLAT
1	XLB	0.0499	State Street Materials Select Sector SP...	Materials	FLAT
21	IGE	0.0497	iShares North American Natural Resource...	Energy	LONG
4	XME	0.0493	State Street SPDR S&P Metals & Mining E...	Materials	LONG
17	SLX	0.0490	VanEck Steel ETF	Materials	LONG
6	MOO	0.0488	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0466	Global X Copper Miners ETF	Materials	LONG
9	MLPX	0.0440	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
5	EMLP	0.0439	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
19	NORW	0.0436	Global X MSCI Norway ETF	Norway Equity	LONG
7	AMLP	0.0416	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
8	MLPA	0.0405	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0360	Global X Uranium ETF	Materials	LONG
20	DBB	0.0328	Invesco DB Base Metals Fund	Industrial Metals	LONG
18	UNG	0.0092	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** LONG since 2026-04-13 (22 days, 0.4%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +10.3%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +8.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-05)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5155
Preferred Stock	7	0.3063
Emerging Markets Fixed Income	4	0.1782

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.050

ticker	rep_wt	name	category	etf_signal
HYG	0.916	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
1	HYG	0.0525	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
9	PFF	0.0480	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0467	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWO	0.0466	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0463	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
20	PCY	0.0457	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	LONG
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	LONG
4	FPE	0.0434	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.764 **Signal:** LONG since 2026-05-06 (5 days, -0.4%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +14.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
VHT	1.103	Vanguard Health Care ETF	Healthcare	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	FLAT
16	FHLC	0.0652	Fidelity MSCI Health Care Index ETF	Healthcare	FLAT
9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0639	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	FLAT
10	IXJ	0.0622	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0620	State Street Health Care Select Sector ...	Healthcare	FLAT
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0595	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0573	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0545	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0542	VanEck Pharmaceutical ETF	Healthcare	FLAT
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0510	ARK Genomic Revolution ETF	Healthcare	LONG
11	IHF	0.0471	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – China Equity

Members: 15 **Var explained:** 0.804 **Signal:** LONG since 2026-04-15 (20 days, 4.5%)

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-02-27
- Current 252d return: +18.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-15 > last sell at 2026-02-27)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0642
Hong Kong Equity	1	0.0592

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.010

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

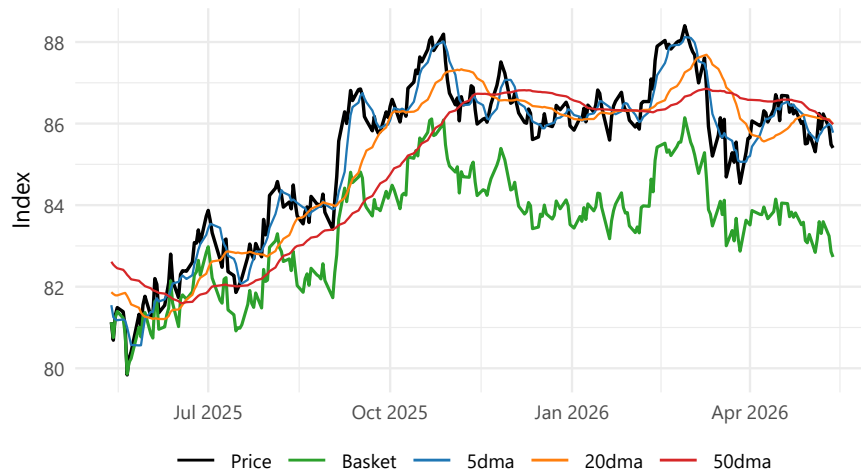
	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0736	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	LONG
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	LONG
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	LONG
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	LONG
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	LONG
3	EMQQ	0.0642	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
10	KBA	0.0594	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
12	EWI	0.0592	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
15	KURE	0.0587	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 13 – Fixed Income

Members: 15 **Var explained:** 0.928 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-02-10 – filter **PASS** (252d at date = +5.7%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +5.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-02-10)



Category breakdown

category	n_members	abs_weight
Investment Grade Corporate Bond	7	0.4706
Fixed Income	6	0.4009
Active Fixed Income	1	0.0665
Taxable Municipal Bond	1	0.0619

Replication basket – TLT (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
TLT	0.693	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
9	BLV	0.0687	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
7	SPLB	0.0678	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
8	IGLB	0.0678	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
10	VCLT	0.0676	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
2	LQD	0.0675	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
11	USIG	0.0672	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
6	QLTA	0.0671	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
12	TLH	0.0667	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
14	SPTL	0.0667	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
15	VGLT	0.0666	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
3	FLCO	0.0665	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
1	TLT	0.0664	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
5	CORP	0.0657	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
13	EDV	0.0657	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
4	BAB	0.0619	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG

Cluster 14 – Real Estate

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (23 days, 3.5%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-03-10
- Current 252d return: +11.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: 0.001

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG

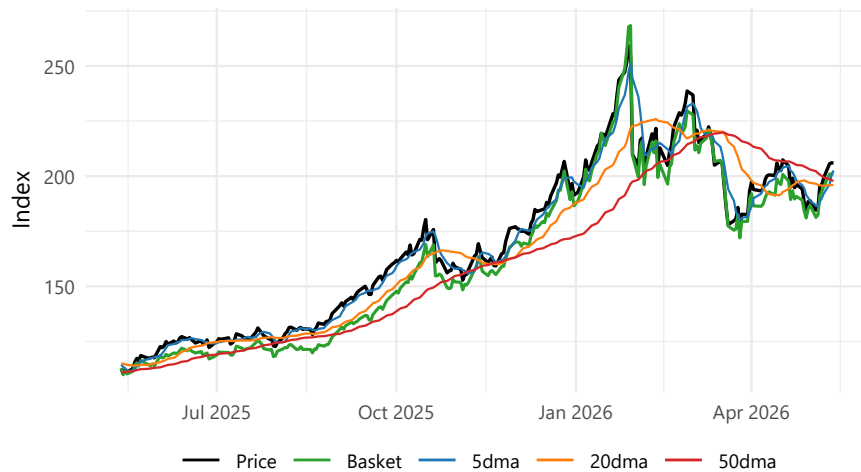
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 15 – Gold

Members: 13 **Var explained:** 0.831 **Signal:** LONG since 2026-05-12 (1 days, -0.0%)

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +76.2%)
- Latest sell_cross: 2026-04-28
- Current 252d return: +83.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-12 > last sell at 2026-04-28)



Category breakdown

category	n_members	abs_weight
Materials	5	0.3889
Gold	3	0.2341
Precious Metals	2	0.1624
Silver	2	0.1524
Platinum	1	0.0622

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
GLD	0.114	SPDR Gold Shares	Gold	FLAT
IAU	0.624	iShares Gold Trust Shares	Gold	FLAT
SLV	0.418	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

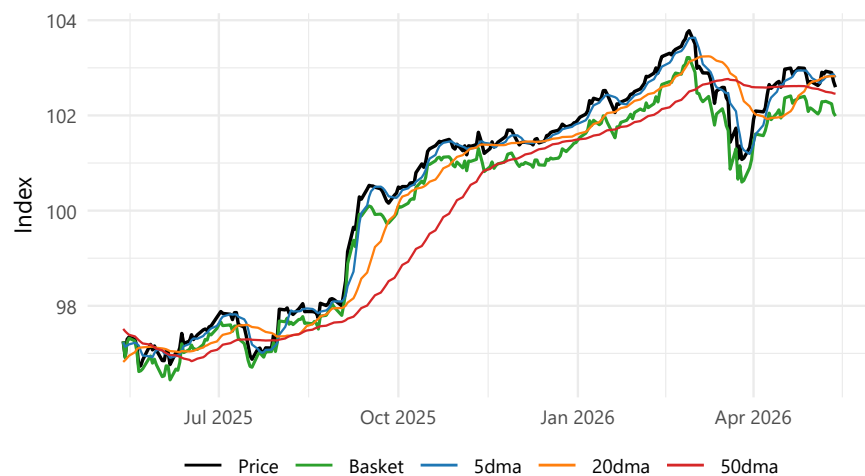
	ticker	PC_wt	name	category	etf_signal
12	DBP	0.0814	Invesco DB Precious Metals Fund	Precious Metals	LONG
7	GLTR	0.0811	abrdn Physical Precious Metals Basket S...	Precious Metals	LONG
2	GDXJ	0.0786	VanEck Junior Gold Miners ETF	Materials	LONG
1	GDX	0.0785	VanEck Gold Miners ETF	Materials	LONG
3	GLD	0.0781	SPDR Gold Shares	Gold	FLAT
10	IAU	0.0781	iShares Gold Trust Shares	Gold	FLAT
11	SGOL	0.0779	abrdn Physical Gold Shares ETF	Gold	FLAT
13	GOAU	0.0777	US Global GO Gold and Precious Metal Mi...	Materials	LONG
6	SIL	0.0776	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0765	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0763	iShares Silver Trust	Silver	LONG
9	SIVR	0.0761	abrdn Physical Silver Shares ETF	Silver	LONG
8	PPLT	0.0622	abrdn Physical Platinum Shares ETF	Platinum	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-14 – filter **PASS** (252d at date = +6.9%)
- Latest sell_cross: 2026-05-05
- Current 252d return: +5.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-05 more recent than last effective buy 2026-04-14)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.182

ticker	rep_wt	name	category	etf_signal
MUB	0.905	iShares National Muni Bond ETF	Municipal Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG

6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0796	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0668	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – India Equity

Members: 12 **Var explained:** 0.420 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-05 – filter **PASS** (252d at date = +12.1%)
- Latest sell_cross: 2026-05-12
- Current 252d return: +8.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-12 more recent than last effective buy 2026-05-05)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.3364
Latin America Equity	1	0.1027
Brazil Equity	1	0.0913
Malaysia Equity	1	0.0904
Chile Equity	1	0.0845
Indonesia Equity	1	0.0831
Philippines Equity	1	0.0778
Vietnam Equity	1	0.0573
Turkey Equity	1	0.0446
Agricultural Commodities	1	0.0319

Replication basket – EWZ+INDA (cor=0.92) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
EWZ	0.256	iShares MSCI Brazil ETF	Brazil Equity	FLAT
INDA	0.513	iShares MSCI India ETF	India Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

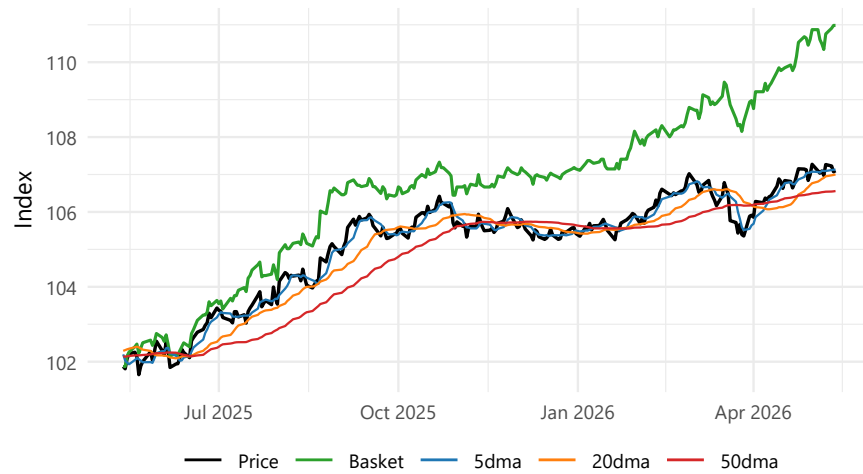
	ticker	PC_wt	name	category	etf_signal
2	INDA	0.1127	iShares MSCI India ETF	India Equity	FLAT
4	INDY	0.1122	iShares India 50 ETF	India Equity	FLAT
5	EPI	0.1114	WisdomTree India Earnings Fund	India Equity	FLAT
7	ILF	0.1027	iShares Latin America 40 ETF	Latin America Equity	FLAT
1	EWZ	0.0913	iShares MSCI Brazil ETF	Brazil Equity	FLAT
10	EWM	0.0904	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	ECH	0.0845	iShares MSCI Chile ETF	Chile Equity	LONG
8	EIDO	0.0831	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
9	EPHE	0.0778	iShares MSCI Philippines ETF	Philippines Equity	FLAT
12	VNM	0.0573	VanEck Vietnam ETF	Vietnam Equity	LONG
11	TUR	0.0446	iShares MSCI Turkey ETF	Turkey Equity	LONG
3	DBA	0.0319	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT

Cluster 18 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.908 **Signal:** LONG since 2026-04-09 (24 days, 0.7%)

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +4.1%)
- Latest sell_cross: 2026-03-20
- Current 252d return: +5.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-09 > last sell at 2026-03-20)

**Category breakdown**

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP (cor=0.90) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
VTIP	1.723	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	LONG
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	LONG
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	LONG
5	TDDT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	LONG
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	LONG
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG

Cluster 19 – Japan Equity

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +39.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7797
Asia-Pacific Equity	2	0.2203

Replication basket – EWJ (cor=0.96) basket lead at lag 1: -0.037

ticker	rep_wt	name	category	etf_signal
EWJ	0.908	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
6	DBJP	0.1115	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1115	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1107	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT

2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1090	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 20 – Broad Commodities

Members: 9 **Var explained:** 0.913 **Signal:** LONG since 2026-01-08 (86 days, 49.5%)

Signal logic:

- Latest buy_cross: 2026-01-08 – filter **PASS** (252d at date = +2.7%)
- Latest sell_cross: 2025-12-15
- Current 252d return: +60.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-01-08 > last sell at 2025-12-15)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6698
Oil	2	0.2183

Energy Commodities	1	0.1119
--------------------	---	--------

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.071

ticker	rep_wt	name	category	etf_signal
PDBC	1.173	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

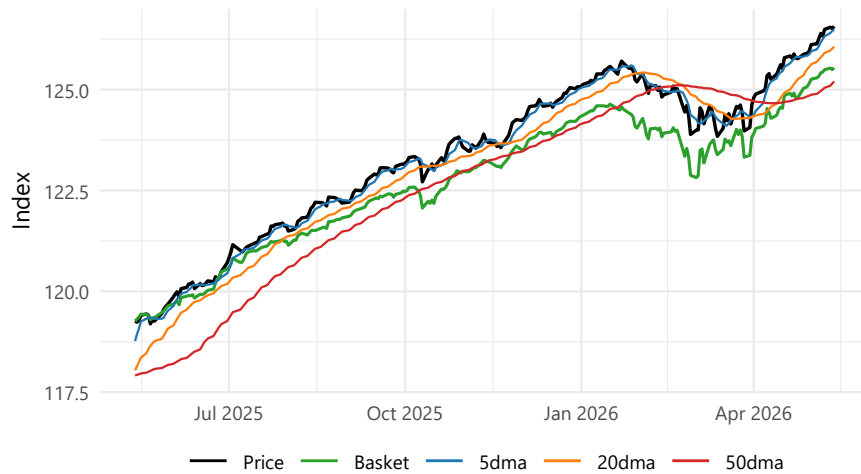
	ticker	PC_wt	name	category	etf_signal
5	GSG	0.1151	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1149	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
7	PDBC	0.1149	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
8	COMT	0.1148	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
9	DBE	0.1119	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.1098	United States Oil Fund	Oil	LONG
4	DBO	0.1085	Invesco DB Oil Fund	Oil	FLAT
6	DJP	0.1051	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	FTGC	0.1050	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT

Cluster 21 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.560 **Signal:** LONG since 2026-04-07 (26 days, 1.2%)

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-02-03
- Current 252d return: +6.3% – filter **ALLOWES** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-02-03)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3632
Interest Rate Hedged Fixed Income	3	0.3245
Floating Rate Fixed Income	3	0.3123

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.048

ticker	rep_wt	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.209	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.311	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.128	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.034	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG

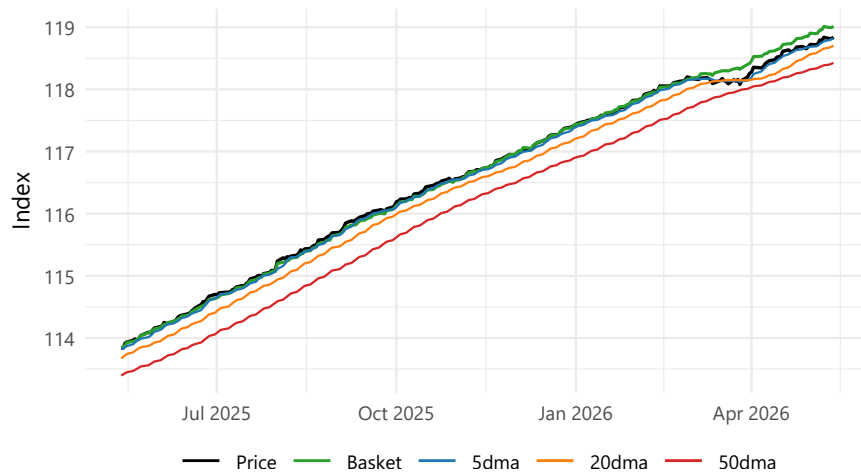
3	BKLN	0.1213	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1170	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1081	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1035	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1007	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 22 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.456 Signal: FLAT

Signal logic:

- Latest buy_cross: 2022-10-31 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-03-24
- Current 252d return: +4.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-03-24 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.108

ticker	rep_wt	name	category	etf_signal
SHV	0.349	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.175	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.117	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.419	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	GSY	0.1359	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1286	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1261	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1244	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1090	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1062	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.1032	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0878	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0787	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 23 – Consumer Staples

Members: 7 **Var explained:** 0.862 **Signal:** LONG since 2026-05-01 (8 days, -0.8%)

Signal logic:

- Latest buy_cross: 2026-05-01 – filter **PASS** (252d at date = +4.7%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +4.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-01 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.8826
U.S. Small Cap Consumer Staples	1	0.1174

Replication basket – XLP (cor=0.97) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
XLP	0.926	State Street Consumer Staples Select Se...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	VDC	0.1520	Vanguard Consumer Staples ETF	Consumer Staples	LONG
5	FSTA	0.1520	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.1503	State Street Consumer Staples Select Se...	Consumer Staples	LONG
2	KXI	0.1459	iShares Global Consumer Staples ETF	Consumer Staples	LONG
3	IYK	0.1453	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
7	PBJ	0.1371	Invesco Food & Beverage ETF	Consumer Staples	FLAT

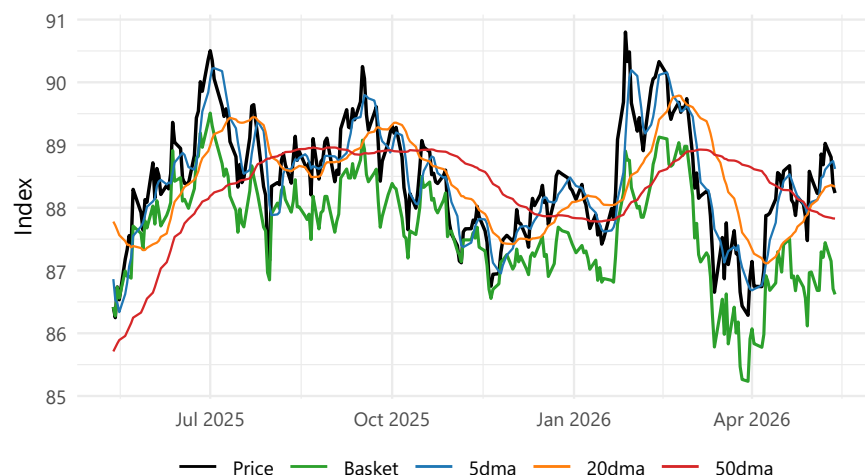
6 PSCC 0.1174 Invesco S&P SmallCap Consumer Staples E... U.S. Small Cap Consumer ... FLAT

Cluster 24 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-04-30 (9 days, -0.4%)

Signal logic:

- Latest buy_cross: 2026-04-30 – filter **PASS** (252d at date = +0.3%)
- Latest sell_cross: 2026-03-03
- Current 252d return: +2.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-30 > last sell at 2026-03-03)



Category breakdown

category	n_members	abs_weight
Currency	4	0.641
International Fixed Income	2	0.359

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.030

ticker	rep_wt	name	category	etf_signal
BWX	0.752	State Street SPDR Bloomberg Internati...	International Fixed I...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	LONG
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	LONG
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1647	Invesco CurrencyShares Swiss Franc Trust	Currency	LONG
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1453	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 25 – Utilities

Members: 4 **Var explained:** 0.997 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-01 – filter **PASS** (252d at date = +19.2%)
- Latest sell_cross: 2026-05-08
- Current 252d return: +13.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-08 more recent than last effective buy 2026-05-01)



Category breakdown

category	n_members	abs_weight
Utilities	4	1

Replication basket – XLU+IDU+VPU+FUTY (cor=1.00) basket lead at lag 1: -0.008

ticker	rep_wt	name	category	etf_signal
XLU	0.247	State Street Utilities Select Sector ...	Utilities	FLAT
IDU	0.252	iShares U.S. Utilities ETF	Utilities	FLAT
VPU	0.255	Vanguard Utilities ETF	Utilities	FLAT
FUTY	0.248	Fidelity MSCI Utilities Index ETF	Utilities	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	XLU	0.2501	State Street Utilities Select Sector SP...	Utilities	FLAT
3	VPU	0.2501	Vanguard Utilities ETF	Utilities	FLAT
4	FUTY	0.2501	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	IDU	0.2497	iShares U.S. Utilities ETF	Utilities	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.624 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-04-08
- Current 252d return: -42.1% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-08 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.018

ticker	rep_wt	name	category	etf_signal
SQQQ	0.507	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.496	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
-----	-------	---------------------------------------	----------	------

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
GBIL	cluster_22	Cash / Ultra-Short Fixe...	2022-08-18	936	17.5%	0.00	1.00
BIL	cluster_22	Cash / Ultra-Short Fixe...	2022-05-19	998	18.0%	0.00	1.00
MINT	cluster_22	Cash / Ultra-Short Fixe...	2023-01-30	824	18.7%	0.00	1.00
GSY	cluster_22	Cash / Ultra-Short Fixe...	2022-12-29	844	19.5%	0.00	1.00
VNLA	cluster_22	Cash / Ultra-Short Fixe...	2026-04-02	28	0.5%	0.06	0.94
SRLN	cluster_21	Bank Loan / Floating Ra...	2026-04-07	26	1.9%	0.00	1.00
LQDH	cluster_21	Bank Loan / Floating Ra...	2026-04-06	27	1.3%	0.00	1.00
IGBH	cluster_21	Bank Loan / Floating Ra...	2026-04-08	25	1.7%	0.00	1.00
FTSL	cluster_21	Bank Loan / Floating Ra...	2026-04-06	27	1.3%	0.01	0.99
BKLN	cluster_21	Bank Loan / Floating Ra...	2026-04-06	27	1.3%	0.01	0.99
IGHG	cluster_21	Bank Loan / Floating Ra...	2026-04-07	26	1.3%	0.09	0.91
XSD	cluster_4	U.S. Large Cap Growth	2026-04-09	24	54.9%	0.00	1.00
PTNQ	cluster_4	U.S. Large Cap Growth	2026-04-17	18	10.2%	0.00	1.00
SMH	cluster_4	U.S. Large Cap Growth	2026-04-08	25	35.4%	0.00	1.00
QTEC	cluster_4	U.S. Large Cap Growth	2026-04-10	23	29.5%	0.00	1.00
FTXL	cluster_4	U.S. Large Cap Growth	2026-04-07	26	56.8%	0.00	1.00
NXTG	cluster_4	U.S. Large Cap Growth	2026-04-08	25	21.1%	0.00	1.00
PSI	cluster_4	U.S. Large Cap Growth	2026-04-07	26	52.7%	0.00	1.00
SOXX	cluster_4	U.S. Large Cap Growth	2026-04-08	25	42.6%	0.00	1.00
IYW	cluster_4	U.S. Large Cap Growth	2026-04-10	23	23.9%	0.00	1.00
QQQ	cluster_4	U.S. Large Cap Growth	2026-04-10	23	17.0%	0.00	1.00
CIBR	cluster_4	U.S. Large Cap Growth	2026-04-17	18	14.1%	0.00	1.00
AIQ	cluster_4	U.S. Large Cap Growth	2026-04-13	22	22.8%	0.00	1.00
XLK	cluster_4	U.S. Large Cap Growth	2026-04-10	23	24.0%	0.00	1.00
XNTK	cluster_4	U.S. Large Cap Growth	2026-04-10	23	23.0%	0.00	1.00
IXN	cluster_4	U.S. Large Cap Growth	2026-04-10	23	23.6%	0.00	1.00
PTLC	cluster_4	U.S. Large Cap Growth	2026-04-20	17	4.8%	0.00	1.00
BOTZ	cluster_4	U.S. Large Cap Growth	2026-04-16	19	13.4%	0.00	1.00
FTEC	cluster_4	U.S. Large Cap Growth	2026-04-10	23	22.1%	0.00	1.00
VGT	cluster_4	U.S. Large Cap Growth	2026-04-10	23	22.1%	0.00	1.00
QTUM	cluster_4	U.S. Large Cap Growth	2026-04-10	23	25.8%	0.00	1.00
IGM	cluster_4	U.S. Large Cap Growth	2026-04-09	24	21.6%	0.00	1.00
ROBO	cluster_4	U.S. Large Cap Growth	2026-04-14	21	14.5%	0.00	1.00
XT	cluster_4	U.S. Large Cap Growth	2026-04-13	22	11.8%	0.00	1.00
ONEQ	cluster_4	U.S. Large Cap Growth	2026-04-13	22	13.8%	0.00	1.00
PSCT	cluster_4	U.S. Large Cap Growth	2026-04-07	26	26.7%	0.00	1.00
QYLD	cluster_4	U.S. Large Cap Growth	2026-04-08	25	4.3%	0.00	1.00

PWB	cluster_4	U.S. Large Cap Growth	2026-04-10	23	13.6%	0.00	1.00
HACK	cluster_4	U.S. Large Cap Growth	2026-04-17	18	8.4%	0.00	1.00
PTF	cluster_4	U.S. Large Cap Growth	2026-04-07	26	31.5%	0.00	1.00
IVW	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.3%	0.00	1.00
VOOG	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.4%	0.00	1.00
SPYG	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.4%	0.00	1.00
QQEW	cluster_4	U.S. Large Cap Growth	2026-04-16	19	6.7%	0.00	1.00
IUSG	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.0%	0.00	1.00
NULG	cluster_4	U.S. Large Cap Growth	2026-04-10	23	14.0%	0.00	1.00
MGK	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.7%	0.00	1.00
ILCG	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.2%	0.00	1.00
ICVT	cluster_4	U.S. Large Cap Growth	2026-04-08	25	10.6%	0.00	1.00
VUG	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.3%	0.00	1.00
EIS	cluster_4	U.S. Large Cap Growth	2026-04-10	23	6.7%	0.00	1.00
TDIV	cluster_4	U.S. Large Cap Growth	2026-04-13	22	12.5%	0.00	1.00
CWB	cluster_4	U.S. Large Cap Growth	2026-04-08	25	10.1%	0.00	1.00
FXL	cluster_4	U.S. Large Cap Growth	2026-04-09	24	15.4%	0.00	1.00
SPHB	cluster_4	U.S. Large Cap Growth	2026-04-10	23	14.6%	0.00	1.00
SCHG	cluster_4	U.S. Large Cap Growth	2026-04-13	22	10.9%	0.00	1.00
PRNT	cluster_4	U.S. Large Cap Growth	2026-04-16	19	9.8%	0.00	1.00
IWY	cluster_4	U.S. Large Cap Growth	2026-04-14	21	8.6%	0.00	1.00
VONG	cluster_4	U.S. Large Cap Growth	2026-04-14	21	8.0%	0.01	0.99
SKYY	cluster_4	U.S. Large Cap Growth	2026-04-17	18	7.2%	0.01	0.99
IWF	cluster_4	U.S. Large Cap Growth	2026-04-14	21	8.0%	0.01	0.99
BLCN	cluster_4	U.S. Large Cap Growth	2026-04-15	20	11.6%	0.01	0.99
IGV	cluster_4	U.S. Large Cap Growth	2026-04-21	16	2.2%	0.01	0.99
ARKW	cluster_4	U.S. Large Cap Growth	2026-04-14	21	11.3%	0.01	0.99
BLOK	cluster_4	U.S. Large Cap Growth	2026-04-13	22	15.3%	0.03	0.97
ARKQ	cluster_4	U.S. Large Cap Growth	2026-04-14	21	10.6%	0.03	0.97
ARKK	cluster_4	U.S. Large Cap Growth	2026-04-14	21	4.3%	0.36	0.64
FDN	cluster_4	U.S. Large Cap Growth	2026-04-10	23	8.9%	0.56	0.44
IPO	cluster_4	U.S. Large Cap Growth	2026-04-13	22	7.8%	0.67	0.33
FINX	cluster_4	U.S. Large Cap Growth	2026-04-20	17	-6.7%	0.74	0.22
EWY	cluster_8	Emerging Markets Equity	2026-04-09	24	35.4%	0.00	1.00
EWT	cluster_8	Emerging Markets Equity	2026-04-08	25	24.2%	0.00	1.00
AIA	cluster_8	Emerging Markets Equity	2026-04-10	23	19.7%	0.00	1.00
AAXJ	cluster_8	Emerging Markets Equity	2026-04-13	22	12.9%	0.00	1.00
EEMA	cluster_8	Emerging Markets Equity	2026-04-13	22	12.7%	0.00	1.00
EMGF	cluster_8	Emerging Markets Equity	2026-04-10	23	12.0%	0.00	1.00
XSOE	cluster_8	Emerging Markets Equity	2026-04-10	23	11.8%	0.00	1.00
EEM	cluster_8	Emerging Markets Equity	2026-04-10	23	11.0%	0.00	1.00
ESGE	cluster_8	Emerging Markets Equity	2026-04-13	22	9.7%	0.00	1.00
GEM	cluster_8	Emerging Markets Equity	2026-04-10	23	10.3%	0.00	1.00
IEMG	cluster_8	Emerging Markets Equity	2026-04-10	23	10.5%	0.00	1.00
EEMV	cluster_8	Emerging Markets Equity	2026-04-10	23	7.7%	0.00	1.00

GMF	cluster_8	Emerging Markets Equity	2026-04-14	21	5.8%	0.00	1.00
DEM	cluster_8	Emerging Markets Equity	2026-04-09	24	5.3%	0.00	1.00
PXH	cluster_8	Emerging Markets Equity	2026-04-10	23	5.2%	0.03	0.97
EWX	cluster_8	Emerging Markets Equity	2026-04-10	23	7.3%	0.09	0.91
SPEM	cluster_8	Emerging Markets Equity	2026-04-13	22	5.1%	0.09	0.91
FNDE	cluster_8	Emerging Markets Equity	2026-04-10	23	4.2%	0.10	0.90
VWO	cluster_8	Emerging Markets Equity	2026-04-13	22	4.8%	0.12	0.88
SCHE	cluster_8	Emerging Markets Equity	2026-04-13	22	4.5%	0.12	0.88
DGS	cluster_8	Emerging Markets Equity	2026-04-10	23	2.7%	0.42	0.58
FEM	cluster_8	Emerging Markets Equity	2026-04-07	26	10.1%	0.50	0.50
SDIV	cluster_8	Emerging Markets Equity	2026-04-09	24	-0.4%	0.93	0.07
VPL	cluster_19	Japan Equity	2026-04-10	23	9.7%	0.00	1.00
BBJP	cluster_19	Japan Equity	2026-04-30	9	4.5%	0.01	0.99
EWJ	cluster_19	Japan Equity	2026-04-13	22	5.4%	0.01	0.99
HEWJ	cluster_19	Japan Equity	2026-04-08	25	4.1%	0.01	0.99
DBJP	cluster_19	Japan Equity	2026-04-08	25	4.1%	0.02	0.98
DXJ	cluster_19	Japan Equity	2026-05-01	8	6.3%	0.02	0.98
EWJV	cluster_19	Japan Equity	2026-05-06	5	2.1%	0.04	0.96
OEF	cluster_5	U.S. Large Cap Equity	2026-04-13	22	10.4%	0.00	1.00
IOO	cluster_5	U.S. Large Cap Equity	2026-04-13	22	10.0%	0.00	1.00
DSI	cluster_5	U.S. Large Cap Equity	2026-04-14	21	8.9%	0.00	1.00
MGC	cluster_5	U.S. Large Cap Equity	2026-04-13	22	9.3%	0.00	1.00
XLG	cluster_5	U.S. Large Cap Equity	2026-04-13	22	10.5%	0.00	1.00
IWL	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.8%	0.00	1.00
VOO	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.2%	0.00	1.00
IVV	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.2%	0.00	1.00
SPY	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.2%	0.00	1.00
ESGU	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.4%	0.00	1.00
OMFL	cluster_5	U.S. Large Cap Equity	2026-04-10	23	7.5%	0.00	1.00
VV	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.2%	0.00	1.00
QAI	cluster_5	U.S. Large Cap Equity	2026-04-08	25	4.0%	0.00	1.00
ILCB	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.2%	0.00	1.00
SUSA	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.1%	0.00	1.00
SPTM	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.7%	0.00	1.00
SPYX	cluster_5	U.S. Large Cap Equity	2026-04-13	22	8.3%	0.00	1.00
SCHX	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.9%	0.00	1.00
IYY	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.8%	0.00	1.00
EPS	cluster_5	U.S. Large Cap Equity	2026-04-10	23	8.7%	0.00	1.00
SCHB	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.7%	0.00	1.00
ITOT	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.8%	0.00	1.00
IWB	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.5%	0.00	1.00
VONE	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.6%	0.00	1.00
VTI	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.7%	0.00	1.00
IWV	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.5%	0.00	1.00
VTHR	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.7%	0.00	1.00

CATH	cluster_5	U.S. Large Cap Equity	2026-04-13	22	7.7%	0.00	1.00
SPHQ	cluster_5	U.S. Large Cap Equity	2026-04-10	23	4.6%	0.00	1.00
ACWI	cluster_5	U.S. Large Cap Equity	2026-04-13	22	6.5%	0.00	1.00
SPGM	cluster_5	U.S. Large Cap Equity	2026-04-10	23	7.2%	0.01	0.99
TILT	cluster_5	U.S. Large Cap Equity	2026-04-13	22	6.1%	0.01	0.99
VT	cluster_5	U.S. Large Cap Equity	2026-04-13	22	6.1%	0.01	0.99
URTH	cluster_5	U.S. Large Cap Equity	2026-04-13	22	6.1%	0.01	0.99
LRGF	cluster_5	U.S. Large Cap Equity	2026-04-13	22	6.7%	0.01	0.99
JHML	cluster_5	U.S. Large Cap Equity	2026-04-13	22	5.8%	0.01	0.99
CRBN	cluster_5	U.S. Large Cap Equity	2026-04-13	22	6.5%	0.01	0.99
AOA	cluster_5	U.S. Large Cap Equity	2026-04-13	22	4.8%	0.01	0.99
GSLC	cluster_5	U.S. Large Cap Equity	2026-04-13	22	6.7%	0.01	0.99
QDF	cluster_5	U.S. Large Cap Equity	2026-04-13	22	5.5%	0.02	0.98
AOR	cluster_5	U.S. Large Cap Equity	2026-04-13	22	3.5%	0.03	0.97
QUAL	cluster_5	U.S. Large Cap Equity	2026-04-14	21	3.8%	0.07	0.93
AOM	cluster_5	U.S. Large Cap Equity	2026-04-13	22	2.3%	0.08	0.92
QUS	cluster_5	U.S. Large Cap Equity	2026-04-14	21	3.0%	0.09	0.91
DGRW	cluster_5	U.S. Large Cap Equity	2026-04-14	21	3.9%	0.13	0.87
AOK	cluster_5	U.S. Large Cap Equity	2026-04-13	22	1.6%	0.14	0.86
FPXI	cluster_5	U.S. Large Cap Equity	2026-04-10	23	7.9%	0.19	0.81
XLY	cluster_5	U.S. Large Cap Equity	2026-04-15	20	0.5%	0.61	0.39
EWS	cluster_5	U.S. Large Cap Equity	2026-04-07	26	3.6%	0.67	0.33
FDIS	cluster_5	U.S. Large Cap Equity	2026-04-14	21	1.7%	0.75	0.25
VCR	cluster_5	U.S. Large Cap Equity	2026-04-14	21	1.7%	0.76	0.24
VOX	cluster_5	U.S. Large Cap Equity	2026-04-13	22	3.8%	0.85	0.15
FCOM	cluster_5	U.S. Large Cap Equity	2026-04-13	22	3.5%	0.87	0.13
XLC	cluster_5	U.S. Large Cap Equity	2026-04-14	21	0.2%	0.86	0.10
IYC	cluster_5	U.S. Large Cap Equity	2026-04-14	21	-1.1%	0.98	-0.01
PDBC	cluster_20	Broad Commodities	2025-12-22	97	40.9%	0.00	1.00
DBC	cluster_20	Broad Commodities	2025-12-22	97	40.1%	0.01	0.99
USO	cluster_20	Broad Commodities	2026-01-29	72	79.5%	0.25	0.74
CQQQ	cluster_12	China Equity	2026-04-20	17	11.0%	0.00	1.00
CNYA	cluster_12	China Equity	2026-04-14	21	6.8%	0.00	1.00
ASHR	cluster_12	China Equity	2026-04-13	22	8.0%	0.00	1.00
KBA	cluster_12	China Equity	2026-04-10	23	12.0%	0.00	1.00
CNXT	cluster_12	China Equity	2026-04-10	23	19.5%	0.00	1.00
EWI	cluster_12	China Equity	2026-04-08	25	3.3%	0.00	1.00
CXSE	cluster_12	China Equity	2026-04-14	21	5.7%	0.06	0.94
FXI	cluster_12	China Equity	2026-04-30	9	4.0%	0.12	0.88
GXC	cluster_12	China Equity	2026-05-04	7	4.4%	0.12	0.88
MCHI	cluster_12	China Equity	2026-05-06	5	1.8%	0.22	0.78
KGRN	cluster_12	China Equity	2026-05-06	5	3.1%	0.41	0.34
SLX	cluster_9	Materials	2026-04-09	24	11.9%	0.00	1.00
AMPL	cluster_9	Materials	2026-04-29	10	0.1%	0.00	0.99
DBB	cluster_9	Materials	2026-04-10	23	8.2%	0.04	0.96

XME	cluster_9	Materials	2026-04-15	20	6.6%	0.05	0.95
COPX	cluster_9	Materials	2026-05-08	3	6.9%	0.08	0.92
NANR	cluster_9	Materials	2026-05-11	2	0.4%	0.13	0.82
MLPA	cluster_9	Materials	2026-04-29	10	-0.1%	0.10	0.73
MLPX	cluster_9	Materials	2026-04-30	9	-2.0%	0.11	0.61
URA	cluster_9	Materials	2026-04-14	21	0.9%	0.84	0.16
NORW	cluster_9	Materials	2025-12-09	106	32.9%	0.78	0.04
IGE	cluster_9	Materials	2026-05-01	8	0.5%	0.41	-0.21
EMLP	cluster_9	Materials	2026-04-28	11	-0.4%	0.73	-0.55
SIVR	cluster_15	Gold	2026-05-11	2	1.8%	0.00	1.00
SLV	cluster_15	Gold	2026-05-11	2	1.7%	0.00	1.00
SILJ	cluster_15	Gold	2026-05-11	2	1.5%	0.05	0.94
GLTR	cluster_15	Gold	2026-05-12	1	0.2%	0.06	0.92
SIL	cluster_15	Gold	2026-05-11	2	0.7%	0.09	0.89
PPLT	cluster_15	Gold	2026-05-11	2	1.3%	0.13	0.82
DBP	cluster_15	Gold	2026-05-12	1	-0.0%	0.14	0.78
GDXJ	cluster_15	Gold	2026-05-12	1	-0.7%	0.25	0.63
GDX	cluster_15	Gold	2026-05-12	1	-0.9%	0.33	0.45
GOAU	cluster_15	Gold	2026-05-13	0	0.0%	0.40	0.24
FXF	cluster_24	International Fixed Inc...	2026-05-05	6	0.1%	0.18	0.82
FXB	cluster_24	International Fixed Inc...	2026-04-13	22	0.3%	0.30	0.70
IGOV	cluster_24	International Fixed Inc...	2026-05-04	7	0.3%	0.45	0.55
BWX	cluster_24	International Fixed Inc...	2026-05-11	2	-0.8%	0.58	0.42
REZ	cluster_14	Real Estate	2026-04-14	21	4.5%	0.00	1.00
NURE	cluster_14	Real Estate	2026-04-13	22	4.8%	0.01	0.99
USRT	cluster_14	Real Estate	2026-04-09	24	4.7%	0.23	0.77
RWR	cluster_14	Real Estate	2026-04-10	23	4.1%	0.26	0.74
PSR	cluster_14	Real Estate	2026-04-09	24	4.3%	0.30	0.70
SCHH	cluster_14	Real Estate	2026-04-09	24	3.9%	0.38	0.62
ICF	cluster_14	Real Estate	2026-04-08	25	4.7%	0.49	0.51
FREL	cluster_14	Real Estate	2026-04-10	23	3.2%	0.51	0.49
XLRE	cluster_14	Real Estate	2026-04-09	24	3.5%	0.54	0.46
RWO	cluster_14	Real Estate	2026-04-13	22	2.4%	0.55	0.45
VNQ	cluster_14	Real Estate	2026-04-10	23	3.1%	0.57	0.43
REET	cluster_14	Real Estate	2026-04-13	22	2.8%	0.60	0.40
INDS	cluster_14	Real Estate	2026-04-14	21	0.2%	0.60	0.40
IYR	cluster_14	Real Estate	2026-04-10	23	3.0%	0.61	0.39
TDTT	cluster_18	Inflation-Protected Fix...	2026-04-01	29	1.2%	0.11	0.89
STPZ	cluster_18	Inflation-Protected Fix...	2026-04-02	28	0.9%	0.14	0.86
TDTF	cluster_18	Inflation-Protected Fix...	2026-04-09	24	0.9%	0.20	0.80
TIPX	cluster_18	Inflation-Protected Fix...	2026-04-07	26	0.8%	0.28	0.72
TIP	cluster_18	Inflation-Protected Fix...	2026-04-10	23	0.6%	0.38	0.62
SCHP	cluster_18	Inflation-Protected Fix...	2026-04-13	22	0.4%	0.46	0.54
SPIP	cluster_18	Inflation-Protected Fix...	2026-04-10	23	0.5%	0.60	0.40
LTPZ	cluster_18	Inflation-Protected Fix...	2026-05-11	2	-0.7%	0.89	-0.36

VIDI	cluster_1	Developed	ex-US	Equity	2026-04-09	24	6.5%	0.01	0.99
ENZL	cluster_1	Developed	ex-US	Equity	2026-05-06	5	-1.4%	0.01	0.99
FNDF	cluster_1	Developed	ex-US	Equity	2026-04-08	25	4.3%	0.02	0.98
PXF	cluster_1	Developed	ex-US	Equity	2026-04-08	25	4.3%	0.02	0.98
IQDF	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.4%	0.03	0.97
HFXI	cluster_1	Developed	ex-US	Equity	2026-04-09	24	5.0%	0.04	0.96
GWX	cluster_1	Developed	ex-US	Equity	2026-04-13	22	3.9%	0.07	0.93
VEU	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.7%	0.07	0.93
SCHF	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.4%	0.08	0.92
FXA	cluster_1	Developed	ex-US	Equity	2026-04-13	22	2.2%	0.08	0.92
SPDW	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.3%	0.08	0.92
VXUS	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.7%	0.09	0.91
VEA	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.3%	0.09	0.91
CWI	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.3%	0.09	0.91
IXUS	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.7%	0.10	0.90
ACWX	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.4%	0.12	0.88
SCZ	cluster_1	Developed	ex-US	Equity	2026-04-10	23	3.3%	0.12	0.88
VSS	cluster_1	Developed	ex-US	Equity	2026-04-13	22	3.8%	0.13	0.87
EWG	cluster_1	Developed	ex-US	Equity	2026-05-05	6	3.5%	0.16	0.84
EWN	cluster_1	Developed	ex-US	Equity	2026-04-10	23	4.6%	0.16	0.84
DLS	cluster_1	Developed	ex-US	Equity	2026-04-13	22	1.8%	0.17	0.83
FNDC	cluster_1	Developed	ex-US	Equity	2026-04-10	23	2.9%	0.20	0.80
IDV	cluster_1	Developed	ex-US	Equity	2026-04-06	27	4.3%	0.21	0.79
IPKW	cluster_1	Developed	ex-US	Equity	2026-04-09	24	1.9%	0.27	0.73
SCHC	cluster_1	Developed	ex-US	Equity	2026-04-13	22	2.5%	0.27	0.73
IMTM	cluster_1	Developed	ex-US	Equity	2026-04-10	23	3.3%	0.30	0.70
FEP	cluster_1	Developed	ex-US	Equity	2026-04-09	24	2.9%	0.32	0.68
EWI	cluster_1	Developed	ex-US	Equity	2026-04-08	25	4.0%	0.33	0.67
VYMI	cluster_1	Developed	ex-US	Equity	2026-04-09	24	1.4%	0.51	0.49
VIGI	cluster_1	Developed	ex-US	Equity	2026-04-14	21	0.5%	0.55	0.45
EW0	cluster_1	Developed	ex-US	Equity	2026-04-10	23	2.9%	0.57	0.43
EZA	cluster_1	Developed	ex-US	Equity	2026-05-08	3	-1.3%	0.53	0.40
BBAX	cluster_1	Developed	ex-US	Equity	2026-04-30	9	0.2%	0.60	0.39
VNQI	cluster_1	Developed	ex-US	Equity	2026-05-04	7	1.3%	0.63	0.37
DWX	cluster_1	Developed	ex-US	Equity	2026-05-04	7	1.0%	0.70	0.29
PID	cluster_1	Developed	ex-US	Equity	2026-04-30	9	-0.3%	0.73	0.26
EWL	cluster_1	Developed	ex-US	Equity	2026-04-15	20	0.2%	0.74	0.26
IHDG	cluster_1	Developed	ex-US	Equity	2026-05-06	5	-1.4%	0.76	0.22
EWG	cluster_1	Developed	ex-US	Equity	2026-04-14	21	-0.1%	0.84	0.16
EMLC	cluster_1	Developed	ex-US	Equity	2026-05-06	5	-0.5%	0.84	0.15
EBND	cluster_1	Developed	ex-US	Equity	2026-05-06	5	-0.8%	0.88	0.11
EWC	cluster_1	Developed	ex-US	Equity	2026-04-09	24	2.3%	0.91	0.09
NFRA	cluster_1	Developed	ex-US	Equity	2026-04-09	24	-0.1%	0.92	0.08
IDLV	cluster_1	Developed	ex-US	Equity	2026-04-30	9	-1.1%	0.96	-0.11
HDEF	cluster_1	Developed	ex-US	Equity	2026-04-07	26	0.9%	0.97	-0.12

EWD	cluster_1	Developed ex-US Equity	2026-05-06	5	-3.6%	0.99	-0.52
IGF	cluster_1	Developed ex-US Equity	2026-04-30	9	-2.1%	0.92	-0.70
VLUE	cluster_2	U.S. Large Cap Value	2026-04-09	24	20.3%	0.00	1.00
IWX	cluster_2	U.S. Large Cap Value	2026-04-10	23	6.7%	0.00	1.00
FNDX	cluster_2	U.S. Large Cap Value	2026-04-09	24	5.9%	0.00	1.00
PRF	cluster_2	U.S. Large Cap Value	2026-04-09	24	6.5%	0.00	1.00
MGV	cluster_2	U.S. Large Cap Value	2026-04-13	22	3.7%	0.00	1.00
IWD	cluster_2	U.S. Large Cap Value	2026-04-10	23	5.8%	0.00	1.00
VONV	cluster_2	U.S. Large Cap Value	2026-04-10	23	5.8%	0.00	1.00
JVAL	cluster_2	U.S. Large Cap Value	2026-04-10	23	8.4%	0.00	1.00
ILCV	cluster_2	U.S. Large Cap Value	2026-04-13	22	4.4%	0.00	1.00
NULV	cluster_2	U.S. Large Cap Value	2026-04-10	23	6.5%	0.00	1.00
RWL	cluster_2	U.S. Large Cap Value	2026-04-10	23	5.7%	0.00	1.00
SCHV	cluster_2	U.S. Large Cap Value	2026-04-10	23	4.3%	0.01	0.99
VTV	cluster_2	U.S. Large Cap Value	2026-04-13	22	2.8%	0.01	0.99
VOOV	cluster_2	U.S. Large Cap Value	2026-04-14	21	3.3%	0.01	0.99
SPYV	cluster_2	U.S. Large Cap Value	2026-04-14	21	3.3%	0.01	0.99
SCHD	cluster_2	U.S. Large Cap Value	2026-04-21	16	2.5%	0.01	0.99
IVE	cluster_2	U.S. Large Cap Value	2026-04-14	21	3.3%	0.01	0.99
IUSV	cluster_2	U.S. Large Cap Value	2026-04-14	21	3.1%	0.04	0.96
RPV	cluster_2	U.S. Large Cap Value	2026-04-20	17	1.2%	0.18	0.82
AIVL	cluster_2	U.S. Large Cap Value	2026-04-10	23	2.8%	0.18	0.82
FEX	cluster_2	U.S. Large Cap Value	2026-04-09	24	3.9%	0.20	0.80
SPYD	cluster_2	U.S. Large Cap Value	2026-04-21	16	0.9%	0.23	0.77
DLN	cluster_2	U.S. Large Cap Value	2026-04-13	22	3.2%	0.24	0.76
DTD	cluster_2	U.S. Large Cap Value	2026-04-13	22	2.9%	0.28	0.72
IMCB	cluster_2	U.S. Large Cap Value	2026-04-10	23	4.9%	0.29	0.71
HDV	cluster_2	U.S. Large Cap Value	2026-05-01	8	-0.6%	0.26	0.67
RDIV	cluster_2	U.S. Large Cap Value	2026-04-21	16	0.2%	0.33	0.67
IWS	cluster_2	U.S. Large Cap Value	2026-04-09	24	3.4%	0.34	0.66
IWR	cluster_2	U.S. Large Cap Value	2026-04-10	23	4.1%	0.37	0.63
DIV	cluster_2	U.S. Large Cap Value	2026-04-07	26	1.1%	0.40	0.60
SIZE	cluster_2	U.S. Large Cap Value	2026-04-13	22	2.9%	0.42	0.58
FDVV	cluster_2	U.S. Large Cap Value	2026-04-14	21	2.7%	0.44	0.56
DIA	cluster_2	U.S. Large Cap Value	2026-04-14	21	2.4%	0.49	0.51
RSP	cluster_2	U.S. Large Cap Value	2026-04-14	21	1.6%	0.51	0.49
ETHO	cluster_2	U.S. Large Cap Value	2026-04-09	24	4.0%	0.58	0.42
VO	cluster_2	U.S. Large Cap Value	2026-04-13	22	2.8%	0.61	0.39
PFM	cluster_2	U.S. Large Cap Value	2026-04-14	21	2.1%	0.64	0.36
VIG	cluster_2	U.S. Large Cap Value	2026-04-14	21	2.1%	0.67	0.33
DHS	cluster_2	U.S. Large Cap Value	2026-04-20	17	0.6%	0.71	0.29
USMV	cluster_2	U.S. Large Cap Value	2026-05-05	6	-0.2%	0.68	0.29
SDOG	cluster_2	U.S. Large Cap Value	2026-04-07	26	1.9%	0.72	0.28
SPHD	cluster_2	U.S. Large Cap Value	2026-05-05	6	-0.8%	0.71	0.24
PWV	cluster_2	U.S. Large Cap Value	2026-04-02	28	4.5%	0.79	0.21

OUSA	cluster_2	U.S. Large Cap Value	2026-04-20	17	-0.7%	0.84	0.16
PEY	cluster_2	U.S. Large Cap Value	2026-04-13	22	0.8%	0.85	0.15
MOAT	cluster_2	U.S. Large Cap Value	2026-05-06	5	-1.5%	0.85	0.14
ACWV	cluster_2	U.S. Large Cap Value	2026-05-06	5	-0.3%	0.75	0.12
DGRO	cluster_2	U.S. Large Cap Value	2026-04-13	22	1.2%	0.91	0.09
VYM	cluster_2	U.S. Large Cap Value	2026-04-10	23	2.4%	0.92	0.08
JPME	cluster_2	U.S. Large Cap Value	2026-04-07	26	3.3%	0.92	0.08
ONEY	cluster_2	U.S. Large Cap Value	2026-04-10	23	2.0%	0.97	0.03
ONEV	cluster_2	U.S. Large Cap Value	2026-04-14	21	0.0%	0.98	0.02
RDVY	cluster_2	U.S. Large Cap Value	2026-04-10	23	2.5%	0.99	0.01
JPUS	cluster_2	U.S. Large Cap Value	2026-04-08	25	0.7%	1.00	0.00
SDY	cluster_2	U.S. Large Cap Value	2026-05-01	8	-1.4%	0.99	-0.03
XLP	cluster_23	Consumer Staples	2026-05-05	6	0.8%	0.00	1.00
IYK	cluster_23	Consumer Staples	2026-05-06	5	1.2%	0.00	1.00
KXI	cluster_23	Consumer Staples	2026-05-06	5	-0.4%	0.06	0.94
VDC	cluster_23	Consumer Staples	2026-05-01	8	-0.2%	0.14	0.86
FSTA	cluster_23	Consumer Staples	2026-05-01	8	-0.2%	0.16	0.84
FV	cluster_3	U.S. Mid Cap Equity	2026-04-14	21	9.3%	0.00	1.00
MTUM	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	17.6%	0.00	1.00
RPG	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	13.4%	0.00	1.00
PRN	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	15.5%	0.00	1.00
PDP	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	9.9%	0.00	1.00
XMMO	cluster_3	U.S. Mid Cap Equity	2026-04-06	27	13.9%	0.00	1.00
FTC	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	9.2%	0.00	1.00
FPX	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	10.6%	0.01	0.99
VFMO	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	10.3%	0.01	0.99
IMCG	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	8.3%	0.04	0.96
QMOM	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	9.1%	0.05	0.95
SCHA	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	7.4%	0.05	0.95
IJK	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	6.1%	0.05	0.95
MDYG	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	6.0%	0.05	0.95
IVOG	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	6.1%	0.06	0.94
SCHM	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	6.1%	0.08	0.92
IWO	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	9.0%	0.08	0.92
VTWG	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	9.1%	0.08	0.92
IWC	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	9.7%	0.10	0.90
IWM	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	8.2%	0.11	0.89
VTWO	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	8.2%	0.11	0.89
VBK	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	8.7%	0.13	0.87
PSC	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	6.7%	0.18	0.82
VOT	cluster_3	U.S. Mid Cap Equity	2026-04-14	21	3.8%	0.19	0.81
ISCG	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	6.2%	0.27	0.73
XMHQ	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	2.8%	0.28	0.72
JETS	cluster_3	U.S. Mid Cap Equity	2026-05-06	5	-4.1%	0.28	0.72
DWAS	cluster_3	U.S. Mid Cap Equity	2026-04-07	26	13.1%	0.36	0.64

SMLF	cluster_3	U.S. Mid Cap Equity	2026-04-09	24	5.2%	0.39	0.61
SPMD	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	3.8%	0.40	0.60
IJH	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	3.8%	0.40	0.60
MDY	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	3.8%	0.40	0.60
IVOO	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	3.8%	0.40	0.60
VB	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	5.0%	0.42	0.58
VXF	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	6.4%	0.46	0.54
IWP	cluster_3	U.S. Mid Cap Equity	2026-04-15	20	0.8%	0.46	0.54
VIS	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	1.2%	0.52	0.48
PAVE	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	3.3%	0.53	0.47
SLYG	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	4.9%	0.54	0.46
FTXR	cluster_3	U.S. Mid Cap Equity	2026-04-17	18	-0.2%	0.54	0.46
IJT	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	5.0%	0.54	0.46
FIDU	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	1.3%	0.55	0.45
VIOG	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	4.8%	0.56	0.44
XLI	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	0.5%	0.59	0.41
XSMO	cluster_3	U.S. Mid Cap Equity	2026-04-08	25	5.7%	0.59	0.41
JHMM	cluster_3	U.S. Mid Cap Equity	2026-04-10	23	3.3%	0.70	0.30
XAR	cluster_3	U.S. Mid Cap Equity	2026-05-12	1	0.6%	0.31	0.29
NUSC	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	2.0%	0.84	0.16
PSCI	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	0.4%	0.89	0.11
IYJ	cluster_3	U.S. Mid Cap Equity	2026-04-13	22	-0.4%	0.90	0.10
IYT	cluster_3	U.S. Mid Cap Equity	2026-04-14	21	0.1%	0.95	0.05
XHB	cluster_3	U.S. Mid Cap Equity	2026-05-06	5	-6.4%	1.00	-0.78
FXD	cluster_3	U.S. Mid Cap Equity	2026-05-06	5	-5.0%	1.00	-0.82
FIW	cluster_3	U.S. Mid Cap Equity	2026-05-08	3	-3.8%	1.00	-0.98
PFXF	cluster_10	High Yield Bond	2026-04-14	21	3.6%	0.00	1.00
PFF	cluster_10	High Yield Bond	2026-04-15	20	1.2%	0.09	0.91
HYEM	cluster_10	High Yield Bond	2026-04-13	22	0.7%	0.27	0.73
VRP	cluster_10	High Yield Bond	2026-04-10	23	0.4%	0.47	0.53
PSK	cluster_10	High Yield Bond	2026-04-17	18	-0.2%	0.69	0.31
PGX	cluster_10	High Yield Bond	2026-04-17	18	-0.7%	0.74	0.26
FPE	cluster_10	High Yield Bond	2026-04-14	21	0.5%	0.85	0.15
HYLS	cluster_10	High Yield Bond	2026-04-09	24	0.4%	0.88	0.12
SHYG	cluster_10	High Yield Bond	2026-04-07	26	0.9%	0.93	0.07
JNK	cluster_10	High Yield Bond	2026-04-08	25	0.2%	0.98	0.02
FALN	cluster_10	High Yield Bond	2026-04-10	23	0.3%	0.98	0.02
HYG	cluster_10	High Yield Bond	2026-04-08	25	0.2%	0.98	0.02
HYS	cluster_10	High Yield Bond	2026-04-08	25	-0.1%	0.99	0.01
BBHY	cluster_10	High Yield Bond	2026-04-08	25	0.3%	0.99	0.01
HYLB	cluster_10	High Yield Bond	2026-04-08	25	0.1%	0.99	0.01
SJNK	cluster_10	High Yield Bond	2026-04-07	26	0.9%	0.99	0.01
HYMB	cluster_16	Municipal Bond	2026-04-09	24	0.2%	0.74	0.26
HYD	cluster_16	Municipal Bond	2026-04-08	25	0.4%	0.78	0.22
ITM	cluster_16	Municipal Bond	2026-05-08	3	-0.3%	0.91	0.08

FMB	cluster_16	Municipal Bond	2026-04-17	18	-0.2%	0.95	0.05
PZA	cluster_16	Municipal Bond	2026-04-10	23	0.1%	0.96	0.04
NYF	cluster_16	Municipal Bond	2026-04-15	20	0.0%	0.97	0.03
MUB	cluster_16	Municipal Bond	2026-04-17	18	-0.4%	0.97	0.03
SUB	cluster_16	Municipal Bond	2026-05-11	2	-0.1%	0.99	-0.28
IHF	cluster_11	Healthcare	2026-05-08	3	3.6%	0.00	1.00
PSCH	cluster_11	Healthcare	2026-04-13	22	5.8%	0.25	0.75
ARKG	cluster_11	Healthcare	2026-04-15	20	-1.0%	0.72	0.27
FXH	cluster_11	Healthcare	2026-04-10	23	2.0%	0.81	0.19
IBB	cluster_11	Healthcare	2026-05-06	5	-0.7%	0.79	-0.02
ISMD	cluster_6	U.S. Small Cap Equity	2026-04-09	24	6.3%	0.19	0.81
OMFS	cluster_6	U.S. Small Cap Equity	2026-04-09	24	5.1%	0.21	0.79
IWN	cluster_6	U.S. Small Cap Equity	2026-04-09	24	4.8%	0.21	0.79
VTWV	cluster_6	U.S. Small Cap Equity	2026-04-09	24	4.7%	0.24	0.76
PRFZ	cluster_6	U.S. Small Cap Equity	2026-04-10	23	5.5%	0.54	0.46
KCE	cluster_6	U.S. Small Cap Equity	2026-04-13	22	4.2%	0.63	0.37
FNDA	cluster_6	U.S. Small Cap Equity	2026-04-10	23	3.9%	0.73	0.27
FNX	cluster_6	U.S. Small Cap Equity	2026-04-10	23	3.0%	0.74	0.26
PSP	cluster_6	U.S. Small Cap Equity	2026-04-15	20	-2.5%	0.76	0.24
EZM	cluster_6	U.S. Small Cap Equity	2026-04-10	23	2.9%	0.78	0.22
VI00	cluster_6	U.S. Small Cap Equity	2026-04-09	24	3.5%	0.79	0.21
IJR	cluster_6	U.S. Small Cap Equity	2026-04-09	24	3.5%	0.80	0.20
SPSM	cluster_6	U.S. Small Cap Equity	2026-04-09	24	3.4%	0.81	0.19
IAI	cluster_6	U.S. Small Cap Equity	2026-04-10	23	5.0%	0.82	0.18
EES	cluster_6	U.S. Small Cap Equity	2026-04-08	25	3.6%	0.85	0.15
FYX	cluster_6	U.S. Small Cap Equity	2026-04-08	25	4.7%	0.86	0.14
VBR	cluster_6	U.S. Small Cap Equity	2026-04-10	23	2.3%	0.87	0.13
RWK	cluster_6	U.S. Small Cap Equity	2026-04-09	24	1.9%	0.90	0.10
IJS	cluster_6	U.S. Small Cap Equity	2026-04-10	23	3.0%	0.96	0.04
SLYV	cluster_6	U.S. Small Cap Equity	2026-04-10	23	3.0%	0.96	0.04
VIOV	cluster_6	U.S. Small Cap Equity	2026-04-10	23	3.1%	0.96	0.04
RWJ	cluster_6	U.S. Small Cap Equity	2026-04-09	24	2.7%	0.98	0.02
RFV	cluster_6	U.S. Small Cap Equity	2026-04-10	23	2.2%	0.99	0.01
DES	cluster_6	U.S. Small Cap Equity	2026-04-08	25	2.0%	0.99	0.01
IJJ	cluster_6	U.S. Small Cap Equity	2026-04-10	23	1.2%	1.00	0.00
DON	cluster_6	U.S. Small Cap Equity	2026-04-13	22	0.2%	1.00	0.00
FNCL	cluster_6	U.S. Small Cap Equity	2026-04-13	22	-1.4%	1.00	-0.01
XLF	cluster_6	U.S. Small Cap Equity	2026-04-13	22	-1.3%	1.00	-0.01
SMMV	cluster_6	U.S. Small Cap Equity	2026-04-09	24	-1.6%	1.00	-0.40
EWM	cluster_17	India Equity	2026-04-22	15	3.5%	0.00	1.00
VNM	cluster_17	India Equity	2026-04-13	22	5.2%	0.16	0.84
TUR	cluster_17	India Equity	2026-04-10	23	0.7%	0.62	0.38
ECH	cluster_17	India Equity	2026-05-06	5	-3.9%	1.00	-0.47
IAGG	cluster_7	Aggregate Fixed Income	2026-05-08	3	-0.5%	0.60	0.34
CMBS	cluster_7	Aggregate Fixed Income	2026-05-08	3	-0.5%	0.71	0.18

SPSB	cluster_7	Aggregate Fixed Income	2026-04-09	24	0.2%	0.83	0.17
LDUR	cluster_7	Aggregate Fixed Income	2026-04-07	26	0.3%	0.84	0.16
BNDX	cluster_7	Aggregate Fixed Income	2026-05-08	3	-0.5%	0.71	0.13
GTO	cluster_7	Aggregate Fixed Income	2026-05-07	4	-0.2%	0.92	-0.02
SPIB	cluster_7	Aggregate Fixed Income	2026-05-06	5	-0.3%	0.96	-0.03
IGIB	cluster_7	Aggregate Fixed Income	2026-05-07	4	-0.1%	0.94	-0.05
VCIT	cluster_7	Aggregate Fixed Income	2026-05-07	4	-0.0%	0.95	-0.10
AGGY	cluster_7	Aggregate Fixed Income	2026-05-08	3	-0.5%	0.95	-0.32
GNMA	cluster_7	Aggregate Fixed Income	2026-05-07	4	-0.5%	0.95	-0.33
VMBS	cluster_7	Aggregate Fixed Income	2026-05-11	2	-0.3%	0.95	-0.46
MBB	cluster_7	Aggregate Fixed Income	2026-05-11	2	-0.3%	0.95	-0.50
SPMB	cluster_7	Aggregate Fixed Income	2026-05-11	2	-0.4%	0.95	-0.68
AGZ	cluster_7	Aggregate Fixed Income	2026-05-11	2	-0.1%	1.00	-0.93
USIG	cluster_13	Fixed Income	2026-05-06	5	-0.4%	0.91	0.03
FLCO	cluster_13	Fixed Income	2026-05-07	4	-0.0%	0.90	0.01
CORP	cluster_13	Fixed Income	2026-05-06	5	-0.5%	0.93	0.00
IGLB	cluster_13	Fixed Income	2026-05-08	3	-0.9%	0.87	-0.01
VCLT	cluster_13	Fixed Income	2026-05-08	3	-0.8%	0.87	-0.03
SPLB	cluster_13	Fixed Income	2026-05-08	3	-0.8%	0.88	-0.07
LQD	cluster_13	Fixed Income	2026-05-08	3	-0.5%	0.91	-0.12
QLTA	cluster_13	Fixed Income	2026-05-08	3	-0.5%	0.95	-0.24
BAB	cluster_13	Fixed Income	2026-05-11	2	-0.5%	0.91	-0.47

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
SMH	cluster_4	U.S. Large Cap Growth	1	1.00
VLUE	cluster_2	U.S. Large Cap Value	1	1.00
IBDR	cluster_7	Aggregate Fixed Income	1	1.00
IHF	cluster_11	Healthcare	1	1.00
MINT	cluster_22	Cash / Ultra-Short Fixed ...	1	1.00
FLTR	cluster_21	Bank Loan / Floating Rate	1	1.00
FV	cluster_3	U.S. Mid Cap Equity	1	1.00
EWY	cluster_8	Emerging Markets Equity	1	1.00
EWM	cluster_17	India Equity	1	1.00
CQQQ	cluster_12	China Equity	1	1.00
OEF	cluster_5	U.S. Large Cap Equity	1	1.00
FTGC	cluster_20	Broad Commodities	1	1.00
VPL	cluster_19	Japan Equity	1	1.00
SLX	cluster_9	Materials	1	1.00
PFXF	cluster_10	High Yield Bond	1	1.00
XLP	cluster_23	Consumer Staples	1	1.00
REZ	cluster_14	Real Estate	1	1.00
SIVR	cluster_15	Gold	1	1.00
VIDI	cluster_1	Developed ex-US Equity	1	0.99
FXY	cluster_24	International Fixed Income	1	0.99

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
IHI	cluster_11	Healthcare	1.00	-1.00
LGLV	cluster_2	U.S. Large Cap Value	1.00	-1.00
SQQQ	cluster_26	Leveraged / Inverse	1.00	-1.00
EIDO	cluster_17	India Equity	1.00	-1.00
IDU	cluster_25	Utilities	1.00	-1.00
PHO	cluster_3	U.S. Mid Cap Equity	1.00	-1.00

IEF	cluster_7	Aggregate Fixed Income	1.00	-0.99
EDV	cluster_13	Fixed Income	0.99	-0.99
PBJ	cluster_23	Consumer Staples	1.00	-0.99
MOO	cluster_9	Materials	0.97	-0.97
EFAV	cluster_1	Developed ex-US Equity	0.94	-0.84
SHM	cluster_16	Municipal Bond	0.99	-0.82
ESPO	cluster_5	U.S. Large Cap Equity	0.99	-0.80
IBUY	cluster_4	U.S. Large Cap Growth	1.00	-0.65
KIE	cluster_6	U.S. Small Cap Equity	1.00	-0.57
LTPZ	cluster_18	Inflation-Protected Fixed...	0.89	-0.36
EMQQ	cluster_12	China Equity	0.98	-0.32
IAU	cluster_15	Gold	0.49	-0.12
ANGL	cluster_10	High Yield Bond	1.00	0.00
SDIV	cluster_8	Emerging Markets Equity	0.93	0.07

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_9	22	FLAT	0.492	0.081	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_22	9	LONG	0.692	0.040	0.00007	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_22	9	LONG	0.635	0.221	0.00035	
FXF	Invesco CurrencyShares Japanese...	Currency	cluster_24	6	FLAT	0.732	0.097	0.00022	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_17	12	FLAT	0.591	0.174	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_22	9	FLAT	0.787	0.131	0.00030	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_22	9	LONG	0.825	0.021	0.00000	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_22	9	FLAT	0.720	0.183	0.00041	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_22	9	FLAT	0.825	0.089	0.00022	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_22	9	LONG	0.656	0.188	0.00047	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_22	9	FLAT	0.793	0.163	0.00038	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_17	12	LONG	0.487	0.220	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_7	43	LONG	0.844	0.125	0.00033	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_22	9	LONG	0.793	0.180	0.00044	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_24	6	LONG	0.818	0.204	0.00048	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_7	43	LONG	0.876	0.161	0.00030	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_7	43	FLAT	0.836	0.232	0.00044	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.877	0.250	0.00044	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_17	12	LONG	0.453	0.334	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.877	0.263	0.00051	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_15	13	LONG	0.751	0.270	0.00073	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_12	15	FLAT	0.804	0.270	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_9	22	LONG	0.751	0.357	0.00083	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_17	12	FLAT	0.515	0.376	0.00106	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_17	12	FLAT	0.580	0.409	0.00106	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_12	15	LONG	0.855	0.254	0.00071	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_21	9	FLAT	0.864	0.332	0.00071	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_18	10	LONG	0.920	0.227	0.00055	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_12	15	LONG	0.905	0.242	0.00063	
ECH	iShares MSCI Chile ETF	Chile Equity	cluster_17	12	LONG	0.637	0.430	0.00117	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20